



Reaching the Centimeter Scale: Optical Detectability Limits for LEO Objects

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Abstract

The growing population of satellites and debris in Low Earth Orbit (LEO) has made space situational awareness a pressing concern, yet the centimeter-scale fragment population lies largely below the detection threshold of operational radar networks.

This thesis develops a first-principles framework for predicting the photometric detectability of LEO targets and their astrometric reference stars from a small ground-based optical system, and applies it to a 8" Schmidt–Cassegrain telescope operating at $f/6.3$ with an IMX432 CMOS detector, operated in the millisecond-exposure synthetic tracking regime from a dark Chilean site. The detection problem for both target classes is reduced to a single photon-counting noise budget, calibrated through a libRadtran atmospheric model, a manufacturer-and-literature-anchored system throughput, and a CALSPEC Vega zero point of $ZP = 20.34 \pm 0.22$ mag. For reference stars, the framework yields stacked limiting magnitudes of $m_{\text{lim}} \approx 14.7\text{--}15.8$ across spectral type, with availability rather than depth binding at high Galactic latitude. For LEO targets, four representative spacecraft surface materials give minimum detectable diameters ranging from ≈ 0.73 cm to ≈ 8.1 cm across the observing envelope; the best case reaches below the radar floor, indicating that a small optical system can in principle access the upper cm-scale under favorable twilight geometry. A first principles parameter-sensitivity analysis quantifies the available hardware upgrades allowing any prospective design change to be evaluated against its detectability gain before hardware is acquired.

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Scope

This thesis develops a quantitative framework for predicting photometric detectability of LEO satellites and their associated reference stars from a small ground-based optical system, and applies it to the specific system used in this work: a Celestron NexStar 8SE Schmidt-Cassegrain telescope operating at $f/6.3$ on an equatorial mount, fitted with a ZWO ASI432MM CMOS detector, and located at a dark Chilean site 1.525 km above sea level. The system is operated in the high-cadence regime, with millisecond-scale sub-exposures recombined through synthetic tracking to recover LEO targets as point sources rather than streaks.

The framework is built around two figures of merit. For astrometric reference stars, the figure of merit is the limiting magnitude $m_{\text{lim}}(T_{\text{eff}}, X)$, the faintest source detectable at $\text{SNR} = 5$ in a sidereal stack at given stellar temperature and airmass. For LEO targets, the figure of merit is the minimum detectable effective cross-section $A_{\text{eff,min}}$, the smallest reflecting area whose reflected-sunlight count rate at the detector exceeds the SNR threshold under given orbital geometry and surface composition. Both figures of merit are derived from the same photon-counting noise budget, specialized to the sidereal and satellite-rate stacks respectively.

Two populations of targets are evaluated. The reference-star analysis is applied to a Gaia DR3 sample restricted to the sky accessible from the observing site with per-star color corrections derived from synthetic photometry against the ATLAS9 stellar grid. The LEO objects analysis is applied to four representative surface materials (multi-layer insulation, white paint, solar-cell coverglass, and protected aluminum) at altitudes spanning $h = 400\text{-}1200$ km and elevations from $\epsilon = 20^\circ$ to 90° .

The scope is restricted in five respects. (i) Targets are treated as unresolved point sources; large satellites whose angular extent exceeds the system PSF are excluded. (ii) Reflectance is modeled as Lambertian and time-averaged over rotation phase; specular glints and phase-resolved light curves are not addressed. (iii) The analysis ends at the detection threshold: orbit determination, identification, and characterization beyond the per-frame photometric magnitude are not within the scope of this work. (iv) No on-sky validation is performed; the throughput is anchored to published end-to-end measurements for comparable instruments, and the detectability limits derived here are predictions rather than measured performance. (v) The framework is deliberately implemented with first-principles models (manufacturer QE, libRadtran atmospheric transmission, ATLAS9 stellar spectra, digitized material reflectances) rather than empirical fits, so that parameter-sensitivity analyses along plausible upgrade paths can be performed without re-calibrating against new observations.

Motivation

Some of the most challenging optical observations from Earth’s surface are those of faint, fast-moving objects and transient events. This regime spans phenomena as diverse as optical pulsars (e.g. Shearer and Golden 2002; Ambrosino et al. 2017), stellar occultations by small bodies such as Kuiper-belt objects (e.g. Schlichting et al. 2009), and near-Earth objects (e.g. Šilha et al. 2018; Karpov et al. 2016). This work focuses on a fourth: artificial satellites and debris in Low Earth Orbit (LEO).

2.1 Space Situational Awareness

Low Earth Orbit hosts the majority of operational spacecraft as well as a growing population of inactive satellites, spent rocket bodies, and fragmentation debris from collisions and breakups. As of the most recent ESA Space Environment Report, approximately 9000 active payloads share orbit with more than 26 000 catalogued debris fragments larger than 10 cm, and statistical models estimate over 10^6 untracked fragments larger than 1 cm [ESA Space Debris Office, 2025]. Atmospheric drag at LEO altitudes is slow enough that objects above a few hundred kilometers can persist for decades or longer, so this population accumulates over time

The foundational paper by Kessler and Cour-Palais [1978] predicted that collisions and debris in LEO would grow exponentially if adequate collision avoidance was not implemented. They demonstrated that once the spatial density of objects exceeds a critical threshold, fragments are generated faster than atmospheric drag can remove them, triggering a self-sustaining cascade of collisions, as shown in Figure 2.1. Thirty

years later, Kessler et al. [2010] confirmed that several altitude bands in LEO already exceeded this threshold under the current catalogued population, and that even 100% compliance with post-mission disposal guidelines would be insufficient to stabilize the environment without the active removal of objects already in orbit.

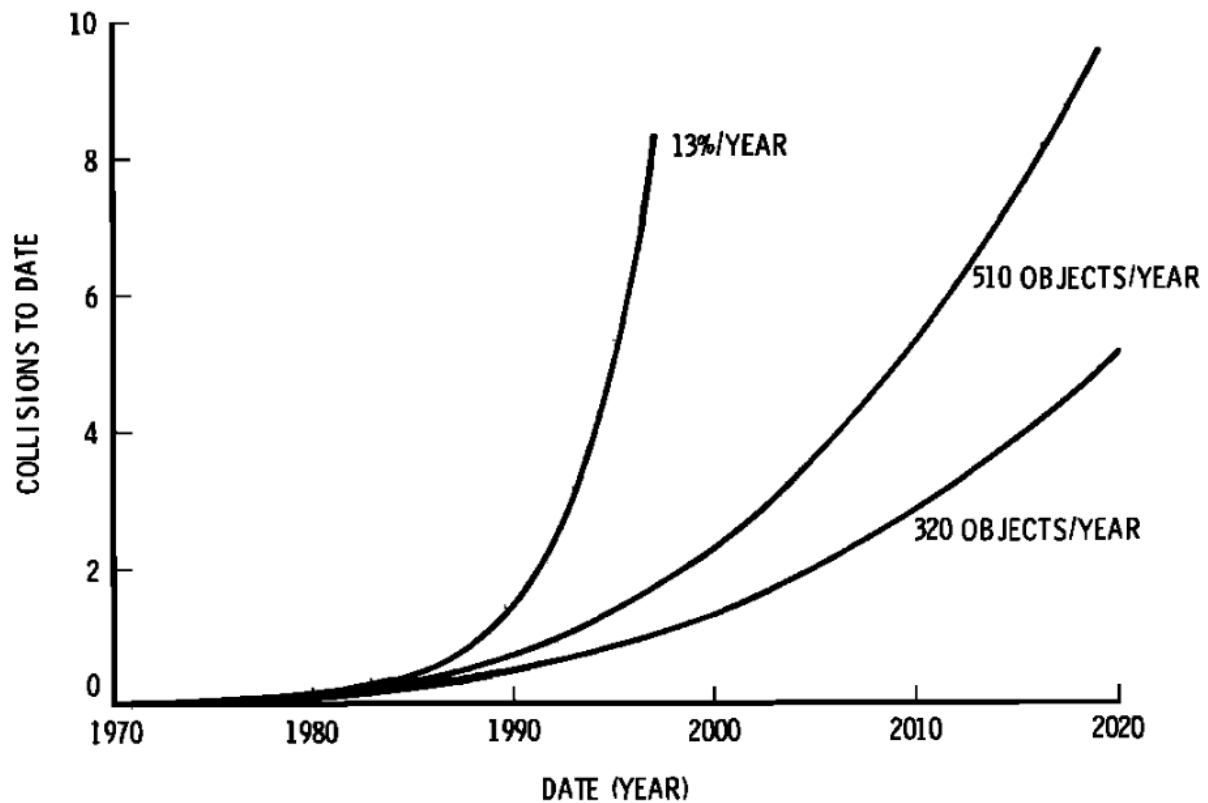


Figure 2.1: Predicted exponential growth of collisions over time, adapted from Kessler and Cour-Palais [1978], under three scenarios for the addition of new objects to LEO: 320 and 510 objects per year and a 13% per-year growth rate.

This is consistent with the findings of the 2025 ESA Space Environment Report [ESA Space Debris Office, 2025], which concludes that even if all new launches were halted immediately, the number of objects in orbit would continue to grow for over 200 years. Figure 2.2 shows the projected evolution of the LEO object population and the cumulative number of catastrophic collisions over the next two centuries under both the current-trend and zero-launch scenarios.

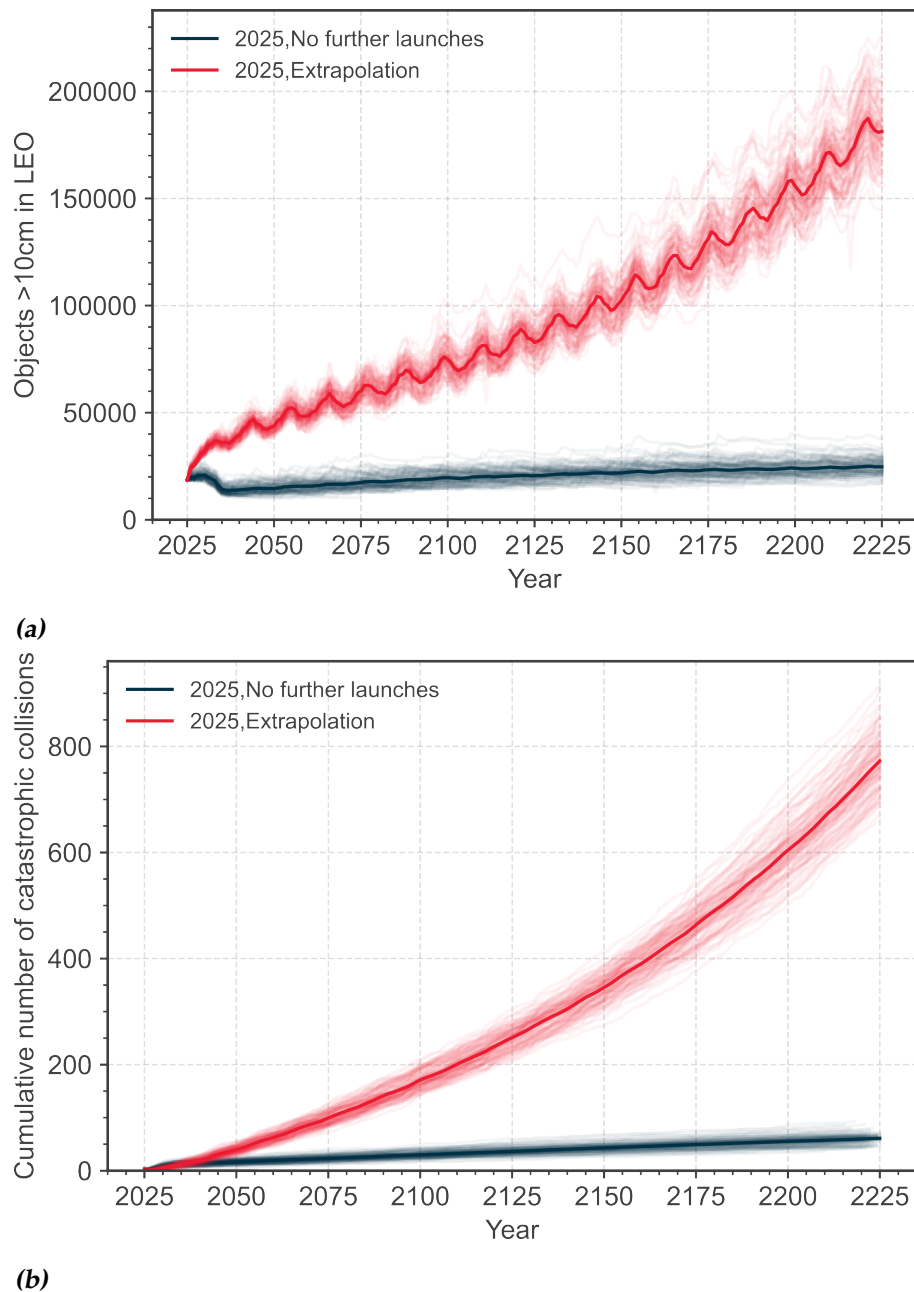
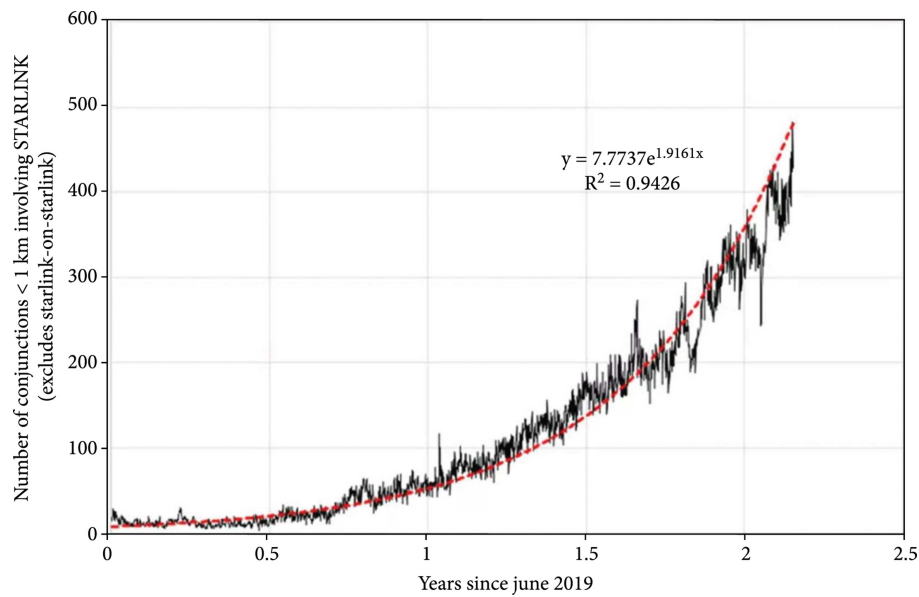


Figure 2.2: Long term projections of the LEO environment, adapted from ESA Space Debris Office [2025]. (a) Number of tracked objects larger than 10 cm in LEO under two scenarios: extrapolation of current launch trends (red)¹ and a zero-launch case in which no further objects are added to orbit (black). (b) Cumulative number of catastrophic collisions in LEO under the same two scenarios.

The rapid deployment of large satellite constellations into LEO in recent years compounds the already unstable orbital environment. With Starlink alone planning a mega-constellation of 42 000 satellites [Zhang et al., 2022], close encounters between

¹The oscillation in is driven by the solar activity cycle acting on atmospheric drag.

operational spacecraft have grown rapidly since the first Starlink launches in 2019 (Figure 2.3a) across a broad range of science cases, an effect that scales directly with the number of objects in orbit and that can only be mitigated through improved orbital knowledge [Lawrence et al., 2022]. Bright constellation satellites also degrade ground-based astronomical observations across a broad range of science cases, contaminating long exposures with satellite trails. Figure 2.3 illustrates both consequences: the rapid growth of close encounters since 2019, and an example of the bright satellite trails that now routinely cross astronomical exposures.



(a)



(b)

Figure 2.3: Two consequences of large LEO constellations. **(a)** The annual number of close encounters between Starlink satellites and other spacecraft since the first Starlink launches in 2019, adapted from Zhang et al. [2022]; the red dashed line is a power-law fit to the data. **(b)** An optical image showing satellite trails from a cluster of Starlink satellites crossing a ground-based exposure, adapted from Lawrence et al. [2022], illustrating the contamination of astronomical observations by bright streaks.

The practical response to this environment is Space Situational Awareness (SSA): the persistent detection, tracking, and characterization of all resident space objects. Current SSA capability rests primarily on three detection methods: ground-based radar networks, laser ranging, and optical telescope networks.

Radar networks are not limited by daylight or cloud cover and provide tracking of objects down to approximately 10 cm in LEO [Haridim et al., 2026]. The most extensive is the Space Surveillance Network (SSN), operated by the United States Space Force.

Satellite laser ranging (SLR) directs short, coherent laser pulses at satellites equipped with retroreflectors; the round-trip time-of-flight gives a distance measurement accurate to millimeters for cooperative targets [Pearlman et al., 2019]. The global SLR network is coordinated by the International Laser Ranging Service, which operates dozens of stations worldwide and supplies data products fundamental to precise orbit determination [Pearlman et al., 2019]. For non-cooperative objects with retroreflectors, diffuse-reflection laser ranging has been demonstrated with ranging precision of order ~ 1 m [Sun et al., 2015] – substantially worse than classical SLR. Optical telescope networks complement radar by having the potential to extend sensitivity to fainter and more distant objects, and by providing photometric and spectroscopic information about object shape, rotation state, and surface properties that radar alone cannot recover [Flohrer et al., 2006].

All three methods face fundamental limitations. Radar networks are limited to objects larger than ~ 10 cm and have geographic coverage gaps that leave portions of each orbit unobserved between station passes [Pearlman et al., 2019]. SLR requires either cooperative retroreflectors or specialized diffuse-reflection techniques that are operationally limited. Optical telescopes are restricted to satellites sunlit against a dark sky, are sensitive to atmospheric seeing and cloud cover, and lose photometric fidelity for fast-moving objects that trail across detector pixels during exposure [Schildknecht, 2007]. The combined consequence is that the centimeter-scale fragment population – large enough to inflict catastrophic damage at orbital velocities, yet below the detection threshold of operational networks – remains effectively invisible [Haridim et al., 2026].

This detection gap has direct operational and regulatory consequences. Satellite operators must perform an increasing number of collision avoidance maneuvers each year, and the reliability of those maneuvers depends directly on the accuracy of the orbital state knowledge of the threatening object [ESA Space Debris Office, 2025]. National space agencies, defense organizations, and regulatory bodies require comprehensive object catalogs to enforce debris mitigation compliance, adjudicate liability

following fragmentation events, and support the long-term governance of the orbital environment [ESA Space Debris Office, 2025]. Improving SSA therefore requires not only broader sensor coverage but also faster cadence, higher photometric precision, and the ability to characterize object properties from optical measurements alone. Fast photometry, the use of high-cadence imaging at millisecond exposure times, is one technique that addresses these needs, and its application to the detection and characterization of LEO satellites and debris forms the subject of this thesis.

Technical Background

3.1 Optical Detectors for High-Speed Imaging

Detector requirements for this work are set by the angular motion of LEO targets across the focal plane. Satellites in LEO travel at orbital velocities of $7.5\text{--}7.8\text{ km s}^{-1}$, which correspond to apparent angular rates approaching $\sim 1^\circ\text{ s}^{-1}$ for near-zenith passes [Vallado, 2013]. At these rates, even an exposure of a few tens of milliseconds smears the satellite signal across many pixels, degrading both the photometric SNR and the astrometric centroid precision.

A simple constraint on the per-frame exposure time follows from requiring that the satellite move no more than a few pixels during a single integration. If the satellite has angular velocity ω and the detector has pixel scale s , the exposure time required to limit trailing to Δx pixels is approximately

$$t_{\text{exp}} \lesssim \frac{\Delta x s}{\omega}, \quad (3.1)$$

with ω and s in matching angular units. For a representative LEO pass at $\omega \sim 1^\circ\text{ s}^{-1}$ and a realistic pixel scale of $1''\text{ pixel}^{-1} = 2.8 \times 10^{-4}\text{ deg pixel}^{-1}$, limiting the per-frame trail to one pixel requires $t_{\text{exp}} \lesssim 0.3\text{ ms}$. Even allowing several pixels of trailing, this scaling places exposure times in the millisecond regime or below, imposing stringent requirement on the detector's noise performance and photon collection efficiency.

At millisecond exposure times, read noise becomes the dominant contribution to the noise budget, exceeding both sky background shot noise and the source shot noise for faint targets. Read noise arises from the on-sensor signal chain: the pixel-level

source follower, reset switching, and the column-level analog-to-digital conversion. It is conventionally expressed in electrons RMS per read. Unlike shot noise, it does not accumulate with exposure time – it sets a fixed noise floor per readout, incurred each time the detector is read out regardless of how short the exposure is. Modern CMOS detectors achieve read noise levels of order $\lesssim 10 e^-$ RMS per read [Karpov et al., 2021], low enough to make high-cadence observations feasible despite the per-frame noise penalty.

Dark current adds a per-pixel signal that scales linearly with exposure time. For modern CMOS sensors at room temperature, dark current is well below $1 e^-$ per pixel per frame at millisecond exposures [Janesick, 2001], and is therefore not a significant noise contributor in this regime. Active cooling, common for long-exposure astronomical imaging, is correspondingly unnecessary for the high-cadence application considered here.

Pixel size influences both photometric sensitivity and the field of view of the system. For a point source imaged through a telescope of fixed focal length, the total photon flux at the detector is independent of pixel size; what changes is how those photons are distributed across pixels. Larger pixels concentrate the source signal into fewer pixels, reducing the read-noise and sky-background contributions to the photometric aperture, both of which scale with the pixel count n_{pix} in the photometric aperture. In the read-noise-dominated regime established above, this reduction directly improves the SNR of individual frames.

Pixel size also sets the angular scale of the detector. For a fixed focal length, larger pixels correspond to a larger plate scale, and the total field of view follows from the plate scale and the array dimensions. For fast-moving LEO targets, a larger FOV translates into a longer dwell time of the satellite on the detector and therefore a larger number of frames available for synthetic tracking (§3.8) [Zhai et al., 2020], which combines short exposures along the satellite’s trajectory to recover the integrated SNR of a tracked observation.

The flexibility of CMOS readout is most useful through region-of-interest (ROI) operation, in which only a sub-window of the sensor is read out each frame. In typical implementations the row count is reduced while the column count is preserved, reflecting the column-parallel architecture: each row still requires one read, but fewer

rows mean a proportionally faster frame cycle. The ROI choice is therefore a direct trade-off between temporal cadence and sky coverage, optimized based on the angular path the target is expected to trace during acquisition.

Two further temporal considerations affect the achievable data quality in the high-cadence regime: the inter-frame dead time and the shutter mechanism. Dead time arises because reading out a frame is not instantaneous; for a typical multi-megapixel CMOS sensor, full-frame readout takes of order tens of milliseconds [Karpov et al., 2021]. Modern architectures mitigate this by overlapping integration with readout so that the next exposure begins while the previous frame is still being read out, reducing the effective dead time to the pixel reset interval – typically tens of microseconds [Meynants, 2009]. The shutter mechanism sets the geometry of each exposure: in rolling shutter operations, rows of the detector are exposed sequentially with each row’s exposure window offset from the next by the inter-row read time. For LEO targets that can cross the detector at several pixels per millisecond, this row-by-row offset distorts the recorded geometry [Liang et al., 2008] and assigns a different epoch to each row, breaking the assumptions that underlie precise astrometry and photometry. A global shutter, in which all pixels begin and end exposure simultaneously, removes both effects (Figure 3.1).

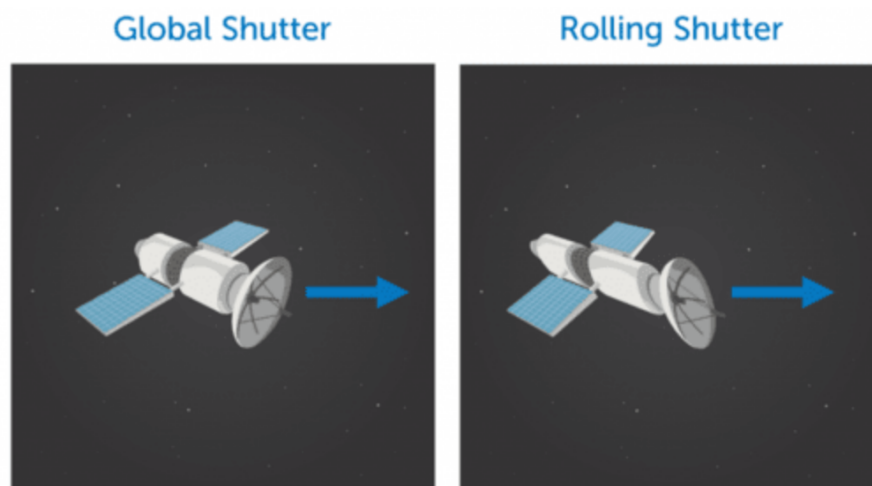


Figure 3.1: Schematic comparison of global (left) and rolling (right) shutter operation, reproduced from Teledyne Vision Solutions [2024]. The satellite appears undistorted with a global shutter but is skewed with a rolling shutter, where each detector row is exposed at a different epoch.

3.2 Schmidt-Cassegrain Telescopes

Schmidt-Cassegrain telescopes (SCTs) are widely used in optical astronomy due to their compact design, relatively fast optics, and versatility across a range of observing applications. Their combination of moderate aperture, manageable physical size, and adaptable focal configurations makes them particularly attractive for high-cadence imaging systems.

A key parameter for imaging applications is the focal ratio. Standard SCTs typically operate at $f/10$, but can be configured to faster systems through reducer optics or front-mounted assemblies. Rear-mounted focal reducers, such as the standard Celestron $0.63\times$ reducer, deliver $f/6.3$ while preserving the rear of the telescope for a standard accessory chain. Front-mounted assemblies such as the Celestron HyperStar replace the secondary mirror with a multi-element corrector lens at the prime focus of the primary mirror, achieving $f/1.9$ on an 8-inch SCT but precluding the use of any rear-mounted accessories.

The rear-mounted configuration accommodates a field rotator, which serves a specific operational role in this work. The column-parallel readout architecture of a modern CMOS detector constrains the available ROI sub-windows to be elongated along one detector axis. A rotator mounted in the rear accessory chain orients the camera once at the start of an observation so the long axis of the ROI matches the satellite's predicted track direction in the detector plane, ensuring that the satellite traverses the entire sub-window rather than crossing it laterally and exiting prematurely.

The system used in this work is mounted on an equatorial mount, which holds the field orientation fixed during integration and therefore allows the rotator to perform a single track-alignment rotation at the start of each observation without further adjustment during the data collection. The fixed field orientation also feeds directly into the synthetic tracking pipeline (§3.8), in which a sidereally tracked image cube is shifted-and-stacked along the satellite's apparent trajectory in post-processing.

The combination of low achievable f-number, compact folded optics, and accessory flexibility makes Schmidt-Cassegrain telescopes well suited to the high-cadence imaging requirements established above. The performance of these optics, however, depends critically on the reflective and transmissive coating applied to their surfaces, discussed in the following subsection.

3.3 Optical Coatings

A telescope's collecting area sets an upper bound on the photon flux delivered to the detector, but the actual throughput depends on how efficiently each optical surface in the light path transmits or reflects the incoming light. For a Schmidt-Cassegrain telescope the relevant surfaces are the corrector plate (two air-glass interfaces) and the primary and secondary mirrors (two reflective surfaces). Each interface absorbs or scatters some fraction of the incident light, and these losses compound along the optical path.

The simplest mirror coating is a single layer of aluminum, which provides broadband reflectivity of $\sim 88\text{--}92\%$ across the visible [Hass and Waylonis, 1961]. Enhanced coatings stack dielectric layers atop the metal to push peak reflectivity above $\sim 96\%$ over a tailored wavelength range [Hass, 1955]. Celestron's StarBright XLT coatings, used on the telescope in this work, follow this approach: a multi-layer dielectric overcoat on the aluminum primary and secondary mirrors, combined with a broadband anti-reflection coating on the corrector plate, give a quoted total system transmission above $\sim 88\%$ over the visible band [Celestron, 2023]. The wavelength dependence of the combined coating performance is one of the dominant terms in the system throughput model developed in §4.1.1 and is discussed quantitatively there.

3.4 Atmospheric effects - libRadtran

The radiative transfer equation governing atmospheric light transport is well established [Chandrasekhar, 1960], but solving it for a realistic atmosphere – with vertical profiles of temperature, pressure, water vapor, ozone, and aerosols, and a spectrally resolved treatment of molecular absorption lines – requires substantial computational machinery. The libRadtran package [Emde et al., 2016, Mayer and Kylling, 2005] provides this capability through a suite of integrated solvers, atmospheric profiles, and absorption parameterizations.

Two libRadtran components are used in this work. Molecular absorption is treated using REPTRAN [Gasteiger et al., 2014], a band-parameterized scheme that reproduces line-by-line accuracy across the optical and near-infrared at substantially reduced computational cost. Radiative transfer is solved with DISORT [Stamnes et al., 1988], the

Discrete-Ordinates Radiative-Transfer algorithm, which handles multiple scattering in a plane-parallel atmosphere and returns transmission for arbitrary viewing geometries. For this thesis, libRadtran is used to generate a pre-computed grid of atmospheric transmission curves $T_{\text{atm}}(\lambda, \epsilon)$, evaluated for a standard mid-latitude summer atmospheric profile at the observing site altitude across a range of elevation angles. The use of these curves in the system throughput model is described in §4.1.1.

3.5 Stellar Spectral Energy Distributions and Photometric Calibration

The count rate that a detector records from an astronomical source depends on how a source's energy output is distributed across wavelength. An observing system — atmosphere, optics, and detector — has a wavelength-dependent transmission, and the source's spectral energy distribution (SED), $F_{\lambda}(\lambda)$, sets how much of its light falls where the system is most sensitive. The atmospheric and instrumental transmission terms are developed in §4.1.1; this subsection establishes the source side.

Photometric calibration that depends on SED shape — such as the color-correction analysis in §4.2.3 — requires SEDs that resolve departures from blackbody behavior. Synthetic spectra computed from model atmospheres provide this: they solve the stellar radiative transfer equation through a vertically structured photosphere with realistic line and continuum opacities, producing flux-calibrated spectra at arbitrary effective temperature, surface gravity, and metallicity. This work uses the ATLAS9 grid [Castelli and Kurucz, 2003], based on plane-parallel hydrostatic LTE models with extensive line blanketing, which is accurate from $T_{\text{eff}} \sim 3500$ K through the OB regime and is the standard choice for the $T_{\text{eff}} \gtrsim 4000$ K stars that dominate any field reference catalog. Stars with $T_{\text{eff}} < 3500$ K, where ATLAS9 under-predicts molecular opacity and more sophisticated grids such as PHOENIX BT-Settl [Allard et al., 2012] would be required, fall below the grid floor and are excluded from the per-star analysis of §5.1.4.

The detected count rate from a source observed through a system of total transmission $\mathcal{T}(\lambda)$ is the photon-counting integral $\dot{C} \propto \int \lambda F_{\lambda}(\lambda) \mathcal{T}(\lambda) d\lambda$. In the Vega magnitude system used throughout this work, the zero-point is fixed by Vega (α Lyr)

itself, and a source's magnitude is

$$m = -2.5 \log_{10} \frac{\dot{C}_\star}{\dot{C}_{\text{Vega}}}, \quad (3.2)$$

with both integrals taken through the same bandpass. The reference SED of Vega is the CALSPEC `alpha_lyr` spectrum [Bohlin et al., 2014], which is flux-calibrated to $\sim 1\%$ across the optical and near-infrared, and the reference bandpass throughout this work is the Bessell [1990] Johnson V filter.

The SED dependence of $\dot{C}$ has a practical consequence: two stars with the same catalogued V magnitude but different T_{eff} deliver different counts through a broad-band, NIR-extended detector. A cool star, whose SED peaks in the red where the CMOS quantum efficiency remains high, delivers more counts per unit V -band flux than Vega; a hot star, whose continuum is nearly Rayleigh–Jeans across the optical, delivers a count rate similar to Vega's. This wavelength-dependent response is represented by a *color correction* that depends on the spectral type of the reference star, and the explicit calculation of this correction for the system used in this work is given in §4.2.3.

3.6 Gaia Catalog

Photometric calibration and astrometric registration of the LEO observations require a reference catalog of well-characterized stars covering the sky accessible from the observing site. The Gaia mission of the European Space Agency [Gaia Collaboration et al., 2016] provides the deepest and most homogeneous such catalog currently available. The third data release, Gaia DR3 [Gaia Collaboration et al., 2016], is the version used throughout this work. Gaia DR3 supplements the astrometric and broadband photometric content of earlier releases with two additional products that are central to the detectability framework. The first is the BP/RP-sampled mean spectra: flux-calibrated, low-resolution spectra spanning $\sim 330\text{--}1050$ nm delivered for the brightest ~ 220 million sources. These spectra are sampled at 343 wavelength points and provide the per-source SED needed for synthetic photometry through any chosen bandpass without requiring model atmospheres.

The second is the Apsis astrophysical-parameter pipeline, which delivers per-source effective temperature T_{eff} , surface gravity $\log g$, metallicity $[M/H]$, and line-of-sight

extinction estimates derived from the BP/RP spectra. The GSP-Phot module produces these parameters by fitting the BP/RP spectra against a grid of synthetic stellar templates; the typical uncertainties at the magnitudes relevant for this work ($G \lesssim 16$) are ~ 170 K in T_{eff} and ~ 0.25 dex in $\log g$ for FGK stars [Andrae et al., 2023b], sufficient for the binary dwarf–giant classification used in the color-correction analysis of §4.2.3.

The Gaia catalog is queried via the Astronomical Data Query Language (AQDL) interface to the Gaia Archive, returning standard photometric and astrometric quantities together with the GSP-Phot stellar parameters required to evaluate per-source detectability in §5.1.4. The reference-star analysis presented in this work is therefore catalog-driven: each query returns a realistic stellar population for a given line of sight, against which the system’s detectability is evaluated on a per-source basis.

3.7 Sensitivity and Detection Thresholds

The figure of merit for a photometric detection is the SNR, the ratio of the source signal in a measurement to the total noise in the same measurement. For a source delivering photon flux Φ above the atmosphere, observed through a system of collecting area A , instrumental throughput $\mathcal{T}_{\text{sys}}$ (combining optical losses and detector quantum efficiency), and atmospheric transmission T_{atm} , the recorded signal in photoelectrons over exposure time t is

$$S = \Phi \mathcal{T}_{\text{sys}} T_{\text{atm}} A_{\text{tel}} t, \quad (3.3)$$

where $\mathcal{T}_{\text{sys}}$ and T_{atm} here denote band-averaged effective values. The wavelength-resolved forms $\mathcal{T}_{\text{sys}}(\lambda)$ and $T_{\text{atm}}(\lambda, \epsilon)$, which depend on wavelength λ and observation elevation ϵ , together with the spectral-dependence of the source flux, enter the photon-counting integral developed in §4.3.

The noise on this signal has four contributions that add in quadrature: shot noise from the source, sky background, dark current, and read noise contributed once per frame by each pixel in the aperture. With $\dot{C}_{\text{sky}}$ the sky count rate per pixel, σ_{rd} the per-pixel read noise, and n_{pix} the number of pixels in the photometric aperture, the total noise is

$$\sigma_{\text{tot}} = \sqrt{S + \dot{C}_{\text{sky}} t n_{\text{pix}} + \dot{C}_{\text{dark}} t n_{\text{pix}} + \sigma_{\text{rd}}^2 n_{\text{pix}}}, \quad (3.4)$$

and the SNR is $\text{SNR} = S/\sigma_{\text{tot}}$.

The dominant noise term depends on the observing regime. In long-exposure deep-sky imaging, sky shot noise grows with t and quickly dominates. In the millisecond-exposure regime considered here, the dark-current contribution is negligible and the read-noise term dominates the noise floor, though the sky-background contribution remains non-negligible and is retained in the full noise budget throughout this work.

A source is considered detected when its SNR exceeds a chosen threshold, conventionally $\text{SNR} \geq 5$ for Gaussian noise [Howell, 2006]. This threshold corresponds to a false-positive probability of $\sim 3 \times 10^{-7}$ per resolution element [Lau et al., 2024], low enough that detections from random fluctuations are not expected across a full image. The faintest source that meets this threshold defines the system’s *limiting magnitude* m_{lim} : solving Eq. 3.4 gives the faintest detectable source for a given exposure time and observing conditions. The analytical limiting magnitude of the system used in this work, derived from the noise budget above, is computed and compared against on-sky measurements in §5.1.

3.8 Synthetic Tracking

The sensitivity framework developed above assumes that the source remains stationary on the detector during an exposure. For LEO targets, this assumption breaks down: at angular rates approaching $\sim 1^\circ \text{s}^{-1}$, even a single-second integration disperses the source signal along a streak hundreds to thousands of pixels long. The total number of source photons collected does not change, but the photometric aperture must enclose the streak, so the per-pixel noise contributions accumulate over an aperture whose size grows linearly with exposure time. The single-exposure approach therefore loses sensitivity faster for faster targets.

Synthetic tracking, also referred to as shifting-and-stacking, addresses this penalty by replacing a single long exposure of duration t_{exp} with a sequence of N short sub-exposures of duration $\tau = t_{\text{exp}}/N$, each chosen short enough that the per-frame streak is comparable to the PSF width [Shao et al., 2014, Zhai et al., 2014]. The sequence is treated as a three-dimensional data cube and shifted along the satellite’s apparent trajectory before summation, so that the LEO target is recovered as a co-aligned point source rather than a streak. The technique was originally developed for asteroid sur-

veys [Shao et al., 2014, Zhai et al., 2014] and has since been applied to faint near-Earth objects with milliarcsecond-level astrometric precision [Zhai et al., 2018].

Replacing a single long exposure with N short sub-exposures changes the noise budget in two ways: read noise accumulates with the number of reads, while sky and dark backgrounds scale with the per-frame streak length rather than the total integration. The sub-exposure time τ therefore becomes a design parameter to be optimized. Section 4.2.1 develops the resulting SNR formalism and the trade-offs that determine the optimal τ for the system used in this work.

3.9 System Summary

The components introduced in the preceding sections combine into the observing system used throughout this work: a Celestron NexStar 8SE Schmidt–Cassegrain telescope on an equatorial mount, configured at $f/6.3$ via a rear-mounted focal reducer, with a derotator and a ZWO ASI432MM CMOS detector. The system is operated at a dark Chilean site at 1.525 km altitude. The full set of system parameters is summarized in Table 3.1. It should also be noted that throughout this work, $n_{pix} = 49$, representing a 7×7 pixel photometric aperture, and the PSF FWHM is 2 pixels.

Telescope	
Model	Celestron NexStar 8SE
Optical design	Schmidt–Cassegrain
Aperture diameter	203.2 mm
Central obstruction	64 mm
Native focal ratio	$f/10$
Focal reducer	Celestron 0.63 $\times$, rear-mounted
Effective focal ratio	$f/6.3$
Optical coatings	StarBright XLT
Field rotator	Rear-mounted derotator
Mount	Equatorial
Detector	
Camera	ZWO ASI432MM
Sensor	Sony IMX432 (monochrome BSI CMOS)
Resolution	1608×1104
Pixel size	$9.0 \mu\text{m}$
Shutter type	Global
Peak quantum efficiency	$\sim 79\%$ near 600 nm
Read noise	$\sim 2.0 \text{ e}^-$ RMS
Full-well capacity	97 ke
Max full-frame rate	120 fps
System geometry	
Plate scale	$1.45''/\text{pixel}$
Full field of view	$38.9' \times 26.7'$
ROI (synthetic tracking)	$38.9' \times 14.5''$

Table 3.1: Summary of the observing system used throughout this work. Detector specifications are drawn from the manufacturer’s product manual [Suzhou ZWO Co., Ltd., 2022]. Optical and coating specifications are drawn from the manufacturer’s published data [Celestron, 2023].

System Characterization and Detectability Framework

The methodology developed in this chapter takes the system introduced in the previous chapter, a Schmidt-Cassegrain telescope on an equatorial mount, fitted with a CMOS detector and operated in a high-cadence, read-noise-dominated regime, and builds the framework used to predict its detectability performance for LEO satellites and reference stars. The framework is developed in three stages. The first establishes the system's wavelength-dependent throughput and photometric zero point through a combination of manufacturer specifications for the optical chain and detector, atmospheric radiative-transfer modeling, and anchoring the absolute normalization to published end-to-end throughput values for comparable ground-based imaging systems (§4.1). The second develops the signal-to-noise framework for synthetic tracking observations of fast-moving targets, including the optimization of the sub-exposure time τ that balances read-noise accumulation against per-frame streak length (§4.2). The third applies the SNR framework to the two target classes: field reference stars used for astrometric calibration (§4.3) and LEO satellites recovered through synthetic tracking (§4.4). The resulting limiting-magnitude expressions and the parameter-sensitivity formalism for the minimum detectable LEO cross-section are applied across a grid of representative observing geometries and surface materials in Chapter 5.

4.1 Calibration

A CMOS detector records electrons per unit time, but these raw counts acquire physical meaning only once the instrument response is known quantitatively. Two quantities are needed: the wavelength-dependent system throughput $\mathcal{T}_{\text{sys}}(\lambda)$ and the photometric zero point ZP .

The system throughput is the probability that a photon arriving at the top of the telescope is ultimately detected as a photoelectron. It combines the reflectivity of the primary and secondary mirrors, the transmission of the corrector plate, and the quantum efficiency of the detector, evaluated on a common wavelength grid. The photon-counting integrals throughout this work convolve $\mathcal{T}_{\text{sys}}(\lambda)$ with the source spectrum and atmospheric transmission to predict the count rate at the detector for a given source.

The photometric zero point converts a detector count rate to an apparent magnitude. With ZP known, any measured count rate $\dot{C}$ corresponds to $m = ZP - 2.5 \log_{10}(\dot{C})$, placing observations on the Vega magnitude system used throughout this work. The zero point follows directly from $\mathcal{T}_{\text{sys}}(\lambda)$ via the photon-counting integral applied to the CALSPEC Vega reference spectrum (§4.1.2).

The remainder of this section describes how $\mathcal{T}_{\text{sys}}(\lambda)$ and ZP were determined. The throughput shape is constructed from manufacturer specifications for the optical chain and detector quantum efficiency (§4.1.1); its absolute normalization is anchored to published end-to-end throughput values for comparable ground-based imaging systems. Atmospheric transmission, treated separately from the instrumental throughput, is modeled with libRadtran and applied per-observation according to the target's elevation angle.

4.1.1 Instrument Throughput and Atmospheric Extinction Model

The system throughput $\mathcal{T}_{\text{sys}}(\lambda)$ is constructed in two stages: a wavelength-dependent *shape* derived from the optical chain components (mirror reflectance, corrector transmission, and detector quantum efficiency), and an absolute *normalization* anchored to published end-to-end throughput values for comparable systems. The wavelength-dependent atmospheric transmission $T_{\text{atm}}(\lambda, \epsilon)$ is modeled separately with libRadtran

and applied per-observation according to the target’s elevation angle in the downstream SNR calculations; it is not incorporated into $\mathcal{T}_{\text{sys}}(\lambda)$ itself.

Detector quantum efficiency. Data was collected throughout this work using a ZWO ASI432MM CMOS camera based on the Sony IMX432 sensor. The detector quantum efficiency $QE(\lambda)$ was digitized from the manufacturer’s product manual [Suzhou ZWO Co., Ltd., 2022] using the WebPlotDigitizer tool [Rohatgi, 2022], then interpolated onto the working wavelength grid. The resulting QE curve is shown in Figure 4.1.

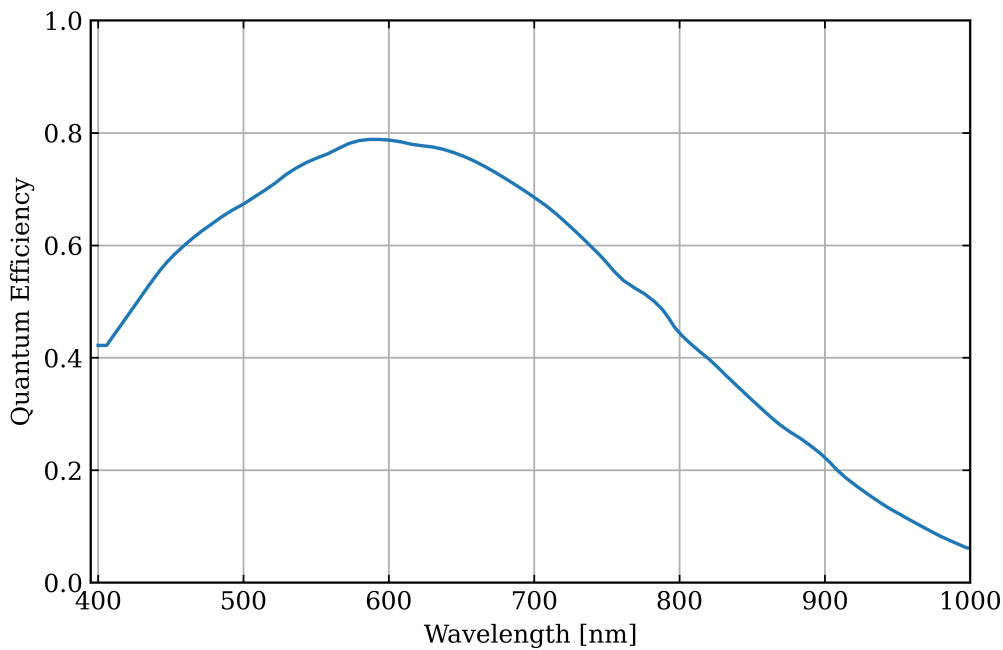


Figure 4.1: Quantum efficiency curve of the ASI432MM CMOS detector digitized from the manufacturer’s product manual [Suzhou ZWO Co., Ltd., 2022].

Optical transmission. The optical chain of the Celestron NexStar 8SE consists of two StarBright XLT-coated mirrors (primary and secondary) and a StarBright XLT-coated corrector plate. Manufacturer-provided transmission data are available only over 450–750 nm [Celestron, 2023]; outside this range a physical model is used to extrapolate to the full detector sensitivity range of 400–1000 nm.

Within the manufacturer-data range, the transmission curve is modeled using a shape-preserving piecewise cubic Hermite interpolator (PCHIP). Outside this range, the system transmission is smoothly blended toward a bare-aluminum reflectance

floor representing the worst-case behavior of a single uncoated reflective surface. The bare-aluminum reflectance $R(\lambda)$ is computed from tabulated optical constants $n(\lambda)$ and $k(\lambda)$ [Palik, 1998, Rakić et al., 1998] using the Fresnel equations at normal incidence. The blend uses cosine-half-window weighting over 70 nm on the blue side (400–450 nm) and 200 nm on the red side (750–1000 nm), with the manufacturer curve held in the central range and the aluminum floor reached at the band edges. The corrector-plate transmission is treated as a wavelength-independent multiplicative factor, justified by the broadband flatness of optical glasses with anti-reflection coating across the visible and near-infrared [Celestron, 2023, Macleod, 2017]. The resulting piecewise function is shown together with the manufacturer data in Figure 4.2.

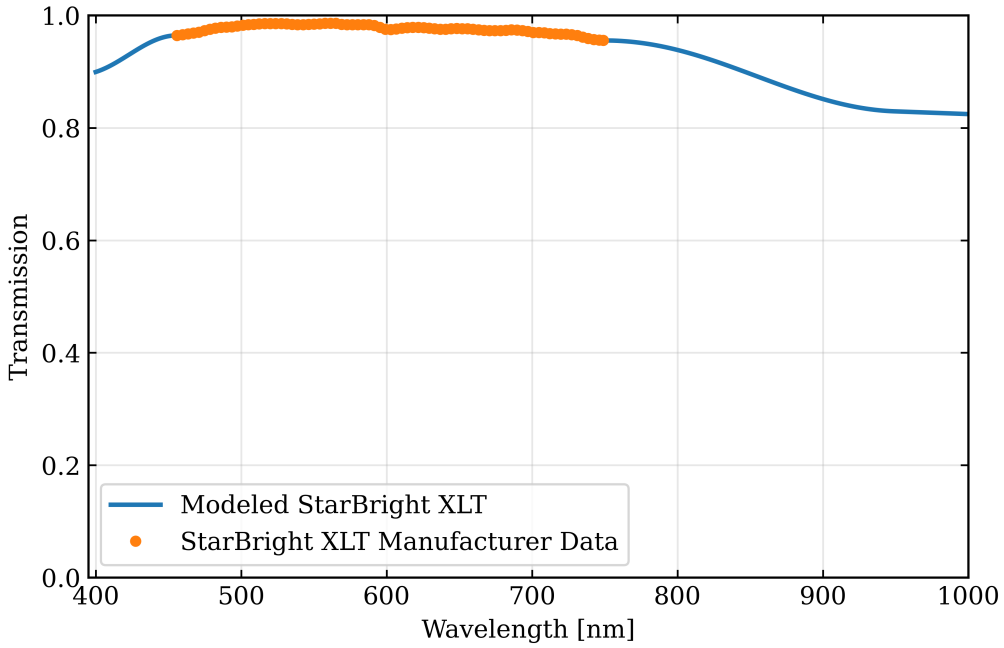


Figure 4.2: Modeled system transmission for the Celestron NexStar 8SE with StarBright XLT coatings. Orange points show the manufacturer data over 450–750 nm [Celestron, 2023]; the blue curve is the piecewise model used throughout this work, extrapolated outside the manufacturer range using a bare-aluminum reflectance floor.

Modeled throughput shape. The wavelength-dependent shape of the system throughput is the product of the optical transmission and the detector QE,

$$\mathcal{T}_{\text{shape}}(\lambda) = T_{\text{opt}}(\lambda) \text{QE}(\lambda), \quad (4.1)$$

and is shown in Figure 4.3. This curve represents the wavelength-dependence of the system response before absolute normalization and before atmospheric attenuation.

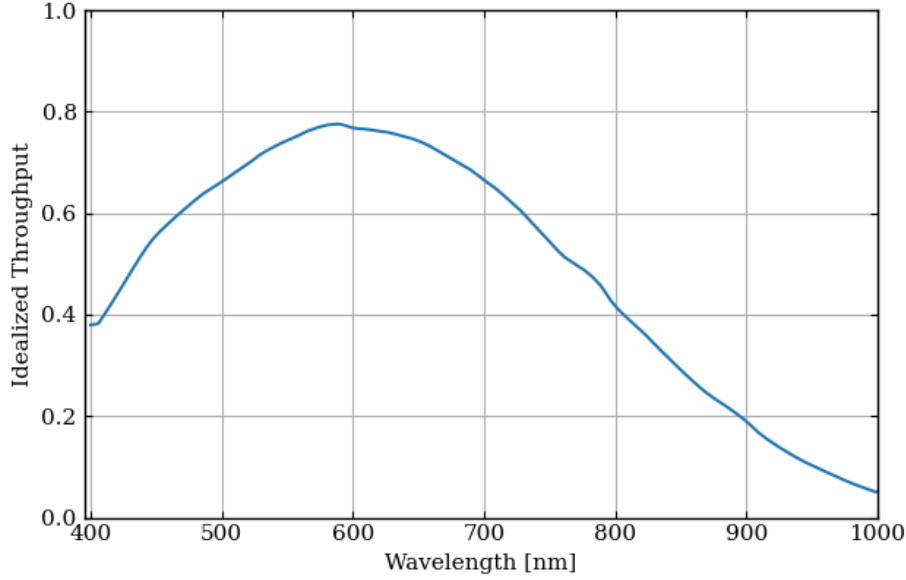


Figure 4.3: Modeled instrument response of the system, computed as the product of the optical transmission of Figure 4.2 and the detector QE of Figure 4.1.

Absolute normalization. The modeled shape $\mathcal{T}_{\text{shape}}(\lambda)$ does not capture the absolute throughput of the system: real ground-based instruments suffer additional losses from mirror coating degradation, dust and surface contamination, imperfect collimation, and aperture-photometry losses that are not captured by the manufacturer component specifications. Rather than deriving the absolute normalization from on-sky standard-star photometry, the modeled throughput is scaled to an absolute peak value anchored to published end-to-end throughput measurement for comparable ground-based optical instruments.

Published measured peak system throughputs for professional optical instruments at high-altitude sites fall in the range $\sim 13\text{--}17\%$, including the Steward Observatory 2.3 m Bok telescope with the Boller & Chivens spectrograph at $\sim 13\%$ peak [Hall, 1995] and the Apache Point Observatory 3.5 m Dual Imaging Spectrograph at $\sim 14\text{--}17\%$ peak [Apache Point Observatory, 1995]. For the 8" Schmidt–Cassegrain system considered here, a peak end-to-end system throughput of 15% is adopted; the detectability predictions in this work are correspondingly normalized to this peak.

A systematic uncertainty of approximately ± 3 percentage points on the adopted peak value reflects the range of plausible end-to-end throughputs for systems comparable to the one used here [Apache Point Observatory, 1995, Hall, 1995]. Following the standard error propagation from fractional flux to magnitude, $\sigma_{\text{mag}} \approx (2.5/\ln 10) \times 3/15$, yields an uncertainty of ~ 0.22 mag that is propagated through all downstream detectability predictions. The scaled system throughput is shown in Figure 4.4.

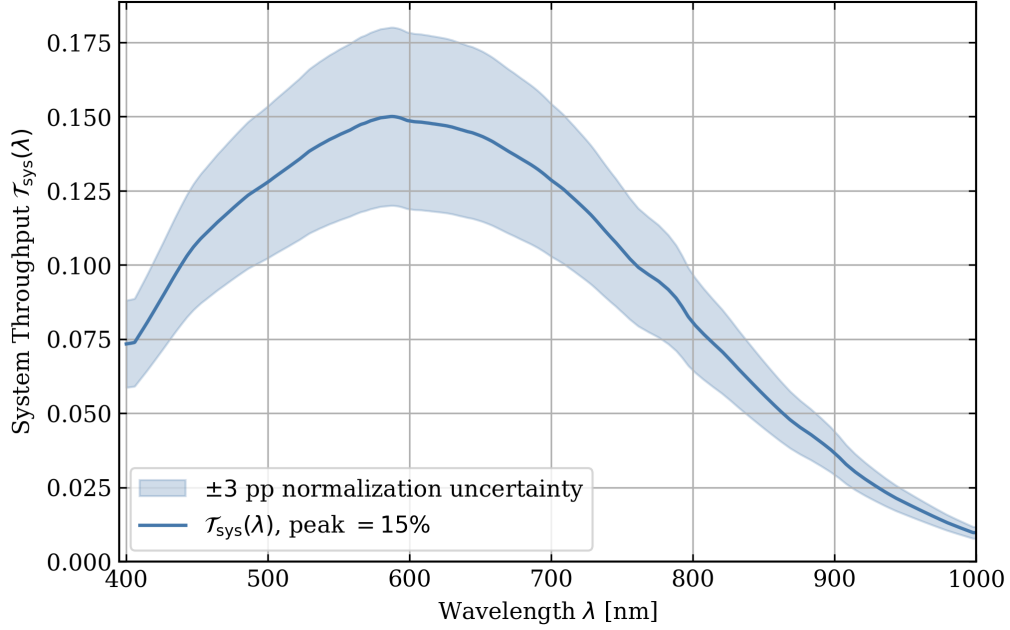


Figure 4.4: System throughput $\mathcal{T}_{\text{sys}}(\lambda)$ used throughout this work, derived by scaling the modeled shape of Figure 4.3 to a peak of 15%. This curve represents the as-built instrumental response before atmospheric attenuation.

Atmospheric transmission. The wavelength-dependent atmospheric transmission $T_{\text{atm}}(\lambda, \epsilon)$ was computed using the libRadtran radiative-transfer package. The uvspec solver was used to generate transmission spectra over the wavelength range 400 – 1000 nm. For each calculation, a standard mid-latitude summer atmospheric profile (midlatitude_summer.dat) was adopted. Molecular absorption was treated using the REPTRAN parameterization, radiative transfer was solved using the DISORT algorithm, and aerosol extinction was incorporated using the default aerosol model. The observer altitude was set to 1.525 km above sea level to approximate the observing site, and transmission was evaluated at ground level ($z_{\text{out}}=0$). For a set of elevation angles ϵ , the wavelength-dependent atmospheric transmission $T_{\text{atm}}(\lambda, \epsilon)$ was computed di-

rectly using the `output_quantity transmittance` option in `uvspec`. The atmospheric transmission for several considered elevation angles is shown in Figure 4.5.

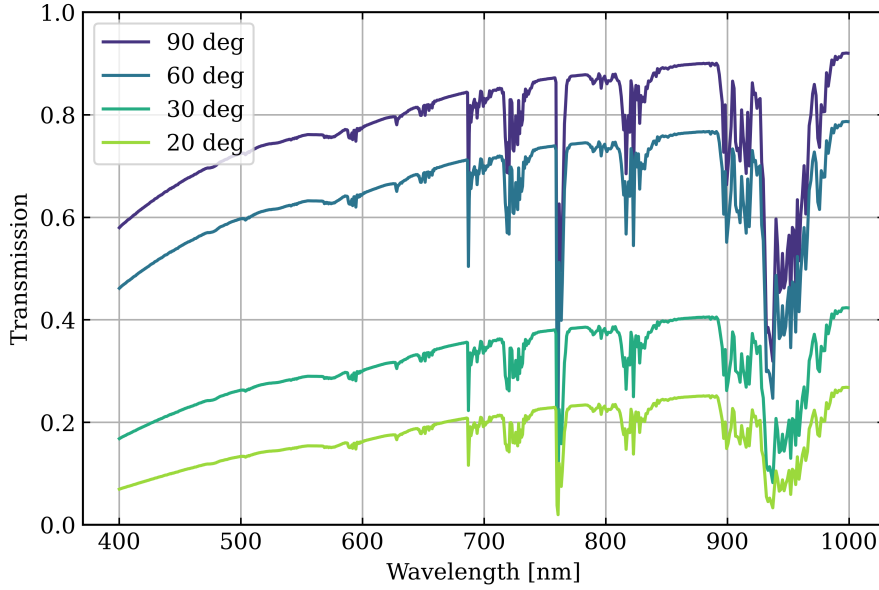


Figure 4.5: Atmospheric transmission $T_{\text{atm}}(\lambda, \epsilon)$ computed with `libRadtran` using `REPTRAN` molecular absorption and the `DISORT` radiative-transfer solver, evaluated for a standard mid-latitude summer profile at the observing site altitude (1.525 km) over elevation angles $\epsilon = 20^\circ - 90^\circ$.

The resulting transmission spectra were assembled into a two-dimensional grid $T_{\text{atm}}(\lambda, \epsilon)$ that was used to interpolate atmospheric transmission at arbitrary observing angles during analysis. Pre-computing this grid allowed the cost of radiative transfer to be incurred only once, enabling more efficient evaluation of atmospheric effects in downstream signal-to-noise and sensitivity calculations.

4.1.2 Zero-point

The photometric zero point ZP of the system is determined by computing the detector count rate from a reference source with a well-defined spectrum, integrated through the empirically scaled system throughput $\mathcal{T}_{\text{sys}}(\lambda)$ from §4.1.1. The reference source is Vega, drawn from the CALSPEC spectral library (`alpha_lyr_stis_011.fits`, Bohlin et al. 2014). The CALSPEC Vega spectrum is a flux-calibrated composite of HST/STIS observations that contains the absorption-line structure of Vega’s atmosphere.

The effective collecting area is computed from the primary mirror diameter (203.2 mm)

after subtracting the area blocked by the central obstruction (64 mm). The Vega count rate at the detector follows from the photon-counting integral

$$\dot{C}_{\text{Vega}} = \frac{A_{\text{tel}}}{hc} \int_0^\infty \lambda F_\lambda^{\text{Vega}}(\lambda) \mathcal{T}_{\text{sys}}(\lambda) d\lambda, \quad (4.2)$$

where $F_\lambda^{\text{Vega}}(\lambda)$ is the CALSPEC reference spectrum. The integral is evaluated by the `synphot` Observation class [STScI Development Team, 2018], which resamples the source spectrum and the system throughput onto a common wavelength grid and computes the integral numerically.

The photometric zero point is then defined as the magnitude of a source producing one electron per second on the detector:

$$ZP = 2.5 \log_{10}(\dot{C}_{\text{Vega}}). \quad (4.3)$$

Evaluating Eq. 4.2 with the scaled throughput $\mathcal{T}_{\text{sys}}(\lambda)$ and the CALSPEC Vega spectrum gives the photometric zero point

$$ZP = 20.34 \pm 0.22 \text{ mag}, \quad (4.4)$$

with the uncertainty propagated from the throughput-scaling spread via $\sigma_{ZP} = (2.5 / \ln 10) \sigma_{\mathcal{T}} / \mathcal{T}_{\text{peak}}$.

4.2 Limiting Magnitude

To determine how faint an object our system can detect, the limiting magnitude is calculated by evaluating the SNR under different observing strategies and conditions. The limiting magnitude is the magnitude of a source whose detected count rate produces exactly the threshold SNR, set throughout this work to $\text{SNR}_{\text{th}} = 5$. Defining $\dot{C}_{\text{lim}}$ as this threshold count rate and ZP as the photometric zero point from §4.1.2,

$$m_{\text{lim}} = ZP - 2.5 \log_{10}(\dot{C}_{\text{lim}}), \quad (4.5)$$

where $\dot{C}_{\text{lim}}$ is obtained by inverting the SNR equation appropriate for the observing strategy. The total noise model includes photon shot noise from both the source and sky background, and per-pixel read noise. Shot noise follows Poisson statistics and therefore scales as the square root of the detected signal, while read noise is treated

as uncorrelated between pixels and frames and added in quadrature during stacking. Dark current is neglected on the basis of the millisecond-exposure analysis in §3.1.

Inverting the noise model for the count rate at which $\text{SNR} = \text{SNR}_{\text{th}}$ yields a quadratic equation in the integrated source counts $C \equiv \dot{C}_{\text{lim}} t_{\text{exp}}$:

$$C^2 - \text{SNR}_{\text{th}}^2 C - \text{SNR}_{\text{th}}^2 \sigma_{\text{bg}}^2 = 0, \quad (4.6)$$

where $\sigma_{\text{bg}}^2 = n_{\text{pix}}(C_{\text{sky}} + N\sigma_{\text{rd}}^2)$ is the total background variance in the photometric aperture over the integration and N is the number of frames stacked. The positive root,

$$C = \frac{1}{2}\text{SNR}_{\text{th}}^2 + \sqrt{\frac{1}{4}\text{SNR}_{\text{th}}^4 + \text{SNR}_{\text{th}}^2 \sigma_{\text{bg}}^2}, \quad (4.7)$$

is the operational definition used throughout this chapter. The remaining task is to specialize the noise model σ_{bg}^2 to the relevant observing strategy: the synthetic-tracking case for the LEO target (§4.2.1), and the sidereal-stack case for reference stars (§4.3.1).

4.2.1 Synthetic Tracking

It is necessary to employ synthetic tracking to maximize the SNR from a LEO target. Extending the Merline & Howell CCD equation (also applicable to CMOS detectors) [Merline and Howell, 1995] to a stack of N frames, the source signal accumulates linearly while the variance contributions add in quadrature per frame, giving

$$\text{SNR}_N = \frac{C_{\star}}{\sqrt{C_{\star} + n_{\text{pix}}C_{\text{sky}} + N n_{\text{pix}} \sigma_{\text{rd}}^2}}, \quad (4.8)$$

where $C_{\star}$ and C_{sky} are the total source and sky counts integrated over the full t_{exp} , and σ_{rd} is the per-pixel read noise of a single frame.

In the background-limited regime, the read-noise term in Eq. 4.8 is negligible and the denominator is dominated by photon noise from the source and sky. The SNR reduces to

$$\text{SNR}_N = \frac{C_{\star}}{\sqrt{C_{\star} + n_{\text{pix}}C_{\text{sky}}}}. \quad (4.9)$$

At fixed sub-exposure time τ , both $C_{\star}$ and the background counts scale linearly with the total integration time $t_{\text{exp}} = N\tau$, so the SNR grows as $\sqrt{N}$ as additional frames are added to the stack. Synthetic tracking incurs no fundamental penalty in this regime:

the per-frame read noise is subdominant, and the result is equivalent to that of a single long exposure of duration t_{exp} at the same limiting magnitude.

In the read-noise dominated regime the variance is instead set by the final term in Eq. 4.8, and the SNR reduces to

$$\text{SNR}_N = \frac{C_\star}{\sqrt{N n_{\text{pix}} \sigma_{\text{rd}}^2}}. \quad (4.10)$$

The behavior with N depends on what is held fixed. Adding frames at fixed τ – and therefore extending the total integration time $t_{\text{exp}} = N\tau$ – increases $C_\star$ linearly with N , while the read-noise variance $N n_{\text{pix}} \sigma_{\text{rd}}^2$ also grows linearly with N . Their ratio in the SNR therefore scales as $N/\sqrt{N} = \sqrt{N}$. Subdividing a fixed total integration into a larger number of shorter frames, by contrast, leaves $C_\star$ unchanged while still accumulating read-noise variance with each additional read, degrading the SNR as $1/\sqrt{N}$ relative to a hypothetical single exposure of the same total duration.

For the system considered in this work, the read-noise term in Eq. 4.8 dominates the noise budget but does not overwhelm it: the sky-background and source-shot-noise contributions remain non-negligible, and all terms are retained in the calculations that follow. The $1/\sqrt{N}$ scaling from subdividing a fixed integration is therefore present but partially diluted relative to the pure read-noise limit. Synthetic tracking is nevertheless the optimal strategy, because the alternative is a single integration of t_{exp} with the signal dispersed across a streak of $\sim \dot{\theta} t_{\text{exp}}/s$ pixels, accumulating sky and read-noise variance over the entire streak aperture. Comparing the two strategies in the read-noise dominated limit, the SNR ratio is

$$\frac{\text{SNR}_{\text{ST}}}{\text{SNR}_{\text{long}}} \approx \sqrt{\ell/w}, \quad (4.11)$$

where $\ell = \dot{\theta} \tau/s$ is the per-frame streak length in pixels with $\dot{\theta}$ the angular velocity of the object, SNR_{ST} is the stacked SNR, and SNR_{long} is the single exposure SNR. Whenever the per-frame streak exceeds the PSF width w , synthetic tracking achieves higher SNR than the streaked single-exposure alternative.

Within the synthetic-tracking strategy, the choice of sub-exposure time τ at fixed t_{exp} is set by two opposing effects. Shortening τ keeps the per-frame streak compact and the photometric aperture small, but increases the number of reads N and therefore the accumulated read noise. Lengthening τ reduces the read-noise contribution but extends the per-frame streak, enlarging the aperture and admitting more sky

background. An optimum sub-exposure time τ_{opt} exists where these two effects are balanced.

To make the optimum quantitative, the streak is treated as approximately rectangular, with length $\dot{\theta} \tau / s$ pixels and cross-track width w pixels set by the PSF, so that the aperture pixel count scales as $n_{\text{pix}} \approx (\dot{\theta} \tau / s) w$. Substituting this into Eq. 4.8 and differentiating with respect to τ yields an optimum of order

$$\tau_{\text{opt}} \sim \left[\frac{w \sigma_{\text{rd}}^2}{(\dot{\theta} / s) \dot{C}_{\text{sky}}} \right]^{1/2}. \quad (4.12)$$

Equation 4.12 should be regarded as an order-of-magnitude estimate; it neglects the detailed shape of the photometric aperture and the structure of the velocity-search grid in the shift-and-stack pipeline, both of which modify the pre-factor at the tens-of-percent level [Zhai et al., 2014]. The scaling, however, is robust: higher read noise extends τ_{opt} to longer exposures, while a brighter sky background or a fast-moving target shortens it. The achievable τ is bounded from below by the sensor's maximum frame rate over the read-out region of interest, and from above by the condition $\dot{\theta} \tau / s \lesssim \text{FWHM}$ that defines the synthetic-tracking regime.

4.2.2 Shift-and-Stack Procedure

The synthetic-tracking observations described above produce a sequence of N short sub-exposures rather than a single integrated frame. This sequence is treated as a three-dimensional data cube $I(x, y, k)$, with two spatial axes and a temporal axis indexing the frame number $k \in \{0, 1, \dots, N-1\}$ [Shao et al., 2014]. The elapsed time at the start of a frame k is $k \cdot \tau$ for sub-exposure duration τ . A stacked image is produced by shifting each frame by a chosen displacement and summing along the temporal axis,

$$S_{\vec{v}}(x, y) = \sum_{k=0}^{N-1} I(x - v_x k \tau, y - v_y k \tau, k), \quad (4.13)$$

where $\vec{v} = (v_x, v_y)$ is the apparent velocity in the image plane (pixels per second) and $v_x k \tau$ is the pixel displacement at frame k . A source whose apparent motion matches $\vec{v}$ is co-aligned across all N frames, and its photons accumulate in the same pixels of $S_{\vec{v}}$; sources moving at any other velocity are smeared into streaks running along their relative motion.

Equation 4.13 can be evaluated repeatedly on the same data cube with different choices of $\vec{v}$, producing distinct stacked images that each isolate a different population of sources. Two such evaluations are relevant for this work. Setting $\vec{v} = 0$ co-aligns sources stationary in the image plane, producing the sidereal stack

$$S_{\text{sid}}(x, y) = \sum_{k=0}^{N-1} I(x, y, k), \quad (4.14)$$

in which field stars appear as point sources at the full integration time t_{exp} , while the satellite traces a streak of length $\theta t_{\text{exp}}/s$ pixels across the field. Setting $\vec{v} = \vec{v}_{\text{sat}}$ to the satellite's apparent velocity instead produces the satellite-rate stack

$$S_{\text{sat}}(x, y) = \sum_{k=0}^{N-1} I(x - v_{x,\text{sat}} k \tau, y - v_{y,\text{sat}} k \tau, k), \quad (4.15)$$

in which the satellite is recovered as a point source at the synthetic-tracking SNR derived in Eq. 4.8 and field stars are smeared anti-parallel to $\vec{v}_{\text{sat}}$.

Both S_{sid} and S_{sat} are derived from the same input cube and therefore share a common pixel coordinate system, and the transformation between them is the known shift $\vec{v}_{\text{sat}} t_{\text{exp}}$ at the cube's reference epoch. The two stacks therefore provide complementary views of the same observation: S_{sat} contains the LEO target as a point source suitable for photometric and astrometric measurement, and S_{sid} contains the field stars at sufficient SNR for astrometric calibration against the Gaia catalog.

A consequence of this construction is that reference stars need not be detectable in individual sub-exposures. The astrometric solution is derived from S_{sid} , in which stars accumulate the full t_{exp} of integration – a $\sqrt{N}$ -deeper image than any single frame in the read-noise dominated regime. The relevant limiting magnitude for reference-star availability is therefore that of the stacked limiting magnitude at t_{exp} , not the per-frame limiting magnitude at τ , and is computed in §5.1.2.

4.2.3 Color Correction

The Vega magnitude system defined in §3.5 calibrates source brightness by comparing the source's photon counts to Vega's, integrated over the same bandpass:

$$m_b = -2.5 \log_{10} \frac{\dot{C}_b^*}{\dot{C}_b^{\text{Vega}}}, \quad (4.16)$$

where $\dot{C}_b^*$ and $\dot{C}_b^{\text{Vega}}$ are the photon count rates delivered by the source and by Vega through the same bandpass b . The reference bandpass used throughout this work is the Johnson V filter, defined by the standardized Bessell V transmission curve [Bessell, 1990]. The choice of V is operational: the Bessell V profile is the standard reference profile for synthetic photometry on the Vega-magnitude system, with well-tabulated absolute calibration against the CALSPEC Vega spectrum [Bohlin et al., 2014]. Both count rates in Eq. 4.16 are evaluated against this transmission curve, and the result is the Vega-magnitude m_V that anchors the Vega-magnitude system. Catalog magnitudes for the reference stars used in this work, drawn from the Gaia DR3 release [Gaia Collaboration et al., 2016], are reported in the Gaia bandpasses G , G_{BP} , and G_{RP} rather than in V , and are converted to the system bandpass through the per-star synthetic transformation derived in §4.3.3.

The system used in this work, however, does not observe through a Bessell V filter. Its effective bandpass is set by the convolution of the optical transmission and the CMOS quantum efficiency, which together define the system response $\mathcal{T}_{\text{sys}}(\lambda)$ derived in §4.1.1. The detected count rate from a star observed through this system is:

$$\dot{C}_{\text{sys}}^* = \frac{A_{\text{tel}}}{hc} \int_{400 \text{ nm}}^{1000 \text{ nm}} \lambda F_{\lambda}^*(\lambda) \mathcal{T}_{\text{sys}}(\lambda) d\lambda. \quad (4.17)$$

Using Eqs. 4.2 and 4.17, the magnitude of a star that would appear in a catalog calibrated to Vega through this system's bandpass is the *system magnitude* m_{sys} :

$$m_{\text{sys}}^* = -2.5 \log_{10} \frac{\dot{C}_{\text{sys}}^*}{\dot{C}_{\text{sys}}^{\text{Vega}}}. \quad (4.18)$$

The V -band magnitude and the system magnitude differ for any star whose SED differs from Vega due to a differing effective temperature T_{eff} , because the Vega-normalized count ratios $\dot{C}_b^* / \dot{C}_b^{\text{Vega}}$ are bandpass-dependent unless the source and Vega share the same spectral shape. The color correction quantifies this offset:

$$\Delta m_{\text{color}}(T_{\text{eff}}) \equiv m_{\text{sys}}^* - m_V^*. \quad (4.19)$$

By construction, $\Delta m_{\text{color}} = 0$ for any source with a Vega-like SED, regardless of its absolute brightness.

The mismatch between the Bessell V reference bandpass and the system bandpass is illustrated in Figure 4.6, together with representative stellar SEDs that show how stars of different temperatures sample the two bandpasses differently.

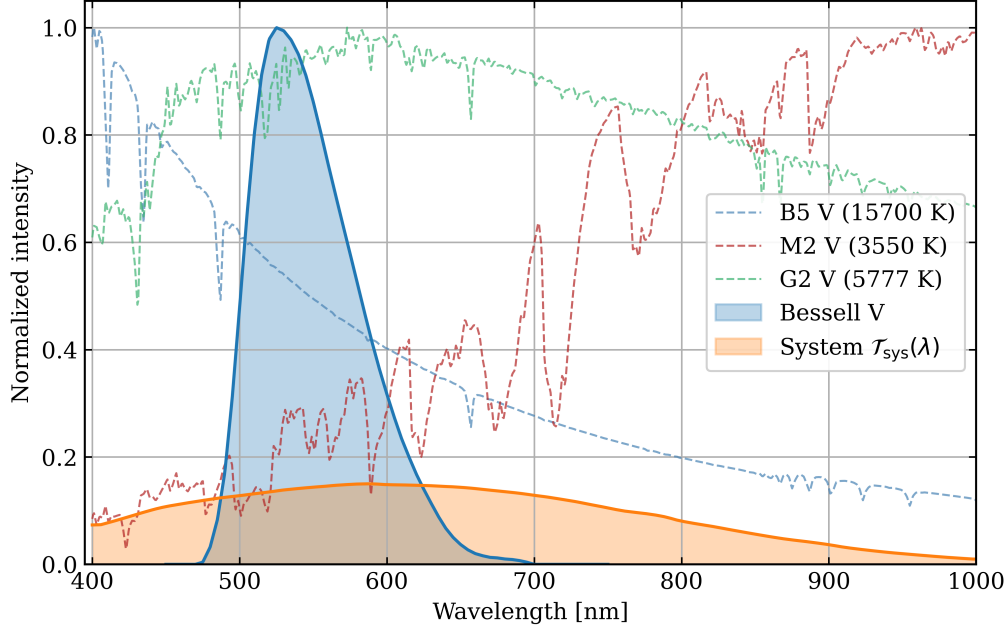


Figure 4.6: Bessell V [Bessell, 1990] and the system bandpass $\mathcal{T}_{\text{sys}}(\lambda)$ derived in §4.1.1, plotted on their natural transmission scale. ATLAS9 [Castelli and Kurucz, 2003] SEDs for a B5 V, G2 V, and M2 V star are overlaid, scaled to fit the same vertical axis for shape comparison; their absolute flux is not represented.

Substituting Eq. 4.18 for m_{sys}^* and the analogous Vega-system definition for m_V^* , and recognizing that the Vega count rate $\dot{C}_b^{\text{Vega}}$ through any bandpass b is a fixed reference quantity, the color correction simplifies to

$$\Delta m_{\text{color}}(T_{\text{eff}}) = -2.5 \log_{10} \left(\frac{\dot{C}_{\text{sys}}^* / \dot{C}_{\text{sys}}^{\text{Vega}}}{\dot{C}_V^* / \dot{C}_V^{\text{Vega}}} \right). \quad (4.20)$$

The ratio inside the logarithm has a natural interpretation: the numerator is the count rate of the star relative to Vega when both are observed through the system bandpass, and the denominator is the same ratio through Bessell V. The color correction therefore measures how the star's brightness relative to Vega differs between the two bandpasses.

The color correction takes a compact form when the stellar SED is normalized to the same V-band flux as Vega. With $F_{\lambda}^*(\lambda)$ scaled so that $\dot{C}_V^* = \dot{C}_V^{\text{Vega}}$ — equivalent to setting $m_V^* = 0$ and asking what the system magnitude would be for a star with Vega's V-band brightness — the denominator of the ratio in Eq. 4.20 becomes unity, and the

expression reduces to

$$\Delta m_{\text{color}}(T_{\text{eff}}) = -2.5 \log_{10} \left(\frac{\dot{C}_{\text{sys}}^{\star}(T_{\text{eff}})}{\dot{C}_{\text{sys}}^{\text{Vega}}} \right). \quad (4.21)$$

The sign convention follows directly from the magnitude scale: a star whose SED concentrates more flux in the system bandpass than Vega has $\dot{C}_{\text{sys}}^{\star} > \dot{C}_{\text{sys}}^{\text{Vega}}$ (at fixed V -band flux), giving $\Delta m_{\text{color}} < 0$ and indicating that the star appears *brighter* in the system bandpass than its m_V would suggest. Conversely, a star whose SED concentrates flux outside the system bandpass has $\Delta m_{\text{color}} > 0$ and appears *fainter* than its m_V . For the system used in this work, whose throughput extends into the near-infrared, cool stars (K and M types) have negative color corrections of magnitude ~ 0.4 – 0.8 , while hot stars (B and A types) have small positive corrections of magnitude $\lesssim 0.01$. The numerical values for the representative spectral classes are tabulated in §5.1.2.

A practical consequence of the color correction is that the limiting magnitude m_{lim} derived for a Vega-like spectrum must be adjusted to predict detectability for stars of other spectral types. Substituting the color-corrected counts into the SNR threshold gives the per-class limiting magnitude

$$m_{\text{lim}}(T_{\text{eff}}) = m_{\text{lim,Vega}} - \Delta m_{\text{color}}(T_{\text{eff}}). \quad (4.22)$$

The color correction enters the per-class limiting magnitude calculation in §4.3.1 (via Eq. 4.26) and is also used to convert Gaia catalog magnitudes to system-bandpass magnitudes for the population analysis framework in §4.3.3.

4.3 Reference Star Detectability

The framework below specializes the photon-counting integral introduced in §4.2.3 to the sidereal-stack detection of reference stars and develops the formalism for evaluating Δm_{color} across realistic reference-star populations. The detected count rate from a source observed at elevation angle ϵ through the system bandpass $\mathcal{T}_{\text{sys}}(\lambda)$ and atmosphere $T_{\text{atm}}(\lambda, \epsilon)$ is

$$\dot{C} = \frac{A_{\text{tel}}}{hc} \int_{400 \text{ nm}}^{1000 \text{ nm}} \lambda F_{\lambda}(\lambda) \mathcal{T}_{\text{sys}}(\lambda) T_{\text{atm}}(\lambda, \epsilon) d\lambda. \quad (4.23)$$

This generalizes Eq. 4.17 by including atmospheric transmission $T_{\text{atm}}(\lambda, \epsilon)$, which was not yet considered in the color-correction derivation. $T_{\text{atm}}(\lambda, \epsilon)$ can be expressed in

terms of the extinction coefficient $k(\lambda)$ and airmass $X(\epsilon)$:

$$T_{\text{atm}}(\lambda, \epsilon) = 10^{-0.4 k(\lambda) X(\epsilon)} \quad (4.24)$$

The reference-star detectability framework is developed in three parts. The SNR formalism is specialized to detection in the sidereal stack in §4.3.1, since reference-star detection happens at the full integration time t_{exp} rather than in individual sub-exposures. The SED-dependent color correction is then computed in two ways: a class-representative tabulation (§4.3.2) that generalizes across spectral types, and a per-star calculation applied to a Gaia DR3 sample (§4.3.3) that predicts the color-correction distribution across realistic fields.

4.3.1 SNR formalism for reference-star detection

An image of a LEO target must contain at least one reference star in the field for astrometric calibration to be possible. As established in §4.2.2, the sidereal stack S_{sid} co-aligns field stars across all N sub-exposures, accumulating their photons over the full integration time t_{exp} . The relevant limiting magnitude for reference-star detectability is therefore the stacked limiting magnitude at t_{exp} , not the per-frame magnitude at τ .

In the read-noise dominated regime, the SNR for a star observed in the sidereal stack follows from Eq. 4.8 with the source held stationary across the cube. The source counts $C_{\star}$ accumulate linearly with the total integration time, while the read-noise variance accumulates as $N n_{\text{pix}} \sigma_{\text{rd}}^2$:

$$\text{SNR}_{\text{sid}} = \frac{C_{\star}}{\sqrt{C_{\star} + n_{\text{pix}} C_{\text{sky}} + N n_{\text{pix}} \sigma_{\text{rd}}^2}}. \quad (4.25)$$

The detected counts $C_{\star}$ follow from the photon-counting integral, Eq. 4.23, evaluated for the appropriate stellar SED and integrated over the full integration time t_{exp} . The detection threshold is set at $\text{SNR}_{\text{sid}} \geq 5$ following the convention introduced in §3.7. Including spectral-type-dependent color correction $\Delta m_{\text{color}}(T_{\text{eff}})$ and the photon-weighted atmospheric extinction $k_{\text{eff}}(T_{\text{eff}})$, the limiting magnitude for a reference star of effective temperature T_{eff} observed at airmass X is

$$m_{\text{lim}}(T_{\text{eff}}, X) = m_{\text{lim, Vega}}^{(t_{\text{exp}})} - \Delta m_{\text{color}}(T_{\text{eff}}) - k_{\text{eff}}(T_{\text{eff}}) X, \quad (4.26)$$

where $m_{\text{lim,Vega}}^{(t_{\text{exp}})}$ is the stacked limiting magnitude for a Vega-like spectrum at the full integration duration. The color-correction term $\Delta m_{\text{color}}(T_{\text{eff}})$ includes the wavelength-dependent response of the system to stars of different spectral types and is defined by Eq. 4.21 in §4.2.3.

The extinction coefficient $k_{\text{eff}}(T_{\text{eff}})$ is the photon-weighted average of the wavelength-dependent extinction $k(\lambda)$ over the bandpass:

$$k_{\text{eff}}(T_{\text{eff}}) = \frac{\int k(\lambda) \Phi_{\lambda}(\lambda) \mathcal{T}_{\text{sys}}(\lambda) d\lambda}{\int \Phi_{\lambda}(\lambda) \mathcal{T}_{\text{sys}}(\lambda) d\lambda}, \quad (4.27)$$

where $\Phi_{\lambda}(\lambda) = F_{\lambda}(\lambda) \lambda / hc$ is the photon arrival rate from the star and $k(\lambda)$ is derived from the libRadtran transmission grid in §4.1.1. The photon-weighting reflects that the detected signal is set by the photon flux convolved with the system response, not by that energy flux integrated against a flat extinction curve. The next two subsections develop the two complementary approaches to evaluating Δm_{color} across the reference-star population.

4.3.2 Class-representative color corrections

The approach to evaluating Δm_{color} across the reference-star population is a class representative tabulation: synthetic SEDs are computed for a small number of representative spectral types spanning the main-sequence and giant branches, and the resulting color corrections are tabulated as a function of spectral class. This generalizes across stars of similar spectral type and provides a look-up-table input to the limiting magnitude calculation that does not require per-star synthetic photometry.

Synthetic SEDs are drawn from the Castelli and Kurucz [2003] ATLAS9 grid of model atmospheres, which provides flux-calibrated continuum spectra for stars across the temperature range relevant for typical reference-star populations. The grid is queried at solar metallicity ($[M/H] = 0$) for all stars. The surface gravity is set by the star's luminosity class: $\log g = 4.5$ (in cgs units of cm s^{-2}) for main sequence dwarfs and $\log g = 2.5$ for giants, with stars classified as dwarfs if $\log g \geq 3.5$ and giants otherwise. The grid is interpolated in effective temperature T_{eff} using the spectral-type-to-temperature scale of Pecaut and Mamajek [2013], with each representative spectral type assigned a T_{eff} from this scale.

The dwarf–giant split is needed because giants have lower surface gravities than dwarfs at the same T_{eff} , and the resulting differences in atmospheric structure produce a small but non-negligible color difference at the system bandpass. Adopting a fixed- $\log g$ treatment for both classes would introduce a systematic color error that varies with the giant fraction of the reference-star population. The fixed-metallicity assumption is appropriate because the metallicity distribution of nearby field stars is centered near solar with a spread of ~ 0.2 dex [Soubiran et al., 2008], well below the precision required for the color correction.

Representative spectral types span B5 V, A0 V, F5 V, G2 V, K5 V, and M2 V on the main sequence (T_{eff} from $\sim 16,000$ to ~ 3500 K), and G8 III, K2 III, and M0 III on the giant branch (T_{eff} from ~ 5000 to ~ 3700 K). Higher-luminosity giants (classes I and II) and the OB regime are excluded as operationally rare in the field reference catalogs at the depths reached by this system. For each representative SED, the color correction Δm_{color} is computed from Eq. 4.21 using the literature-anchored system throughput from §4.1.1. The resulting tabulated corrections, together with the per-class limiting magnitudes derived from Eq. 4.26, appear in §5.1.2.

4.3.3 Population analysis on real fields

Reference stars are queried from Gaia DR3 for representative fields chosen to span the range of Galactic latitudes accessible from the observing site. Six fields are queried, two each at three latitude bands: low-latitude ($b \approx -10^\circ$ and -15°), mid-latitude ($b \approx -42^\circ$ and -52°), and high latitude ($b \approx -62^\circ$ and -76°). The low-latitude fields sample the disc-dominated regime where reference stars are abundant; the high-latitude fields sample the sparsely populated regime where reference-star yield is the limiting factor for astrometric calibration; and the mid-latitude fields bridge the two. The two-fields-per-band design provides a basic measure of field-to-field variation at fixed latitude, in addition to the latitude dependence of the stellar density.

Each catalogued star carries an effective temperature T_{eff} (from the `teff_gspphot` column) and a surface gravity $\log g$ (from `logg_gspphot`), both derived from Gaia’s BP/RP spectra by the GSP-Phot module of the Apsis pipeline [Andrae et al., 2023a]. Andrae et al. [2023a] report typical differences from literature values of ~ 110 K in T_{eff} and ~ 0.2 – 0.25 dex in $\log g$ for the FGK stars at the magnitudes relevant here, sufficient

for the binary dwarf–giant classification used in the color-correction calculation. Stars with $\log g \geq 3.5$ are queried from the ATLAS9 grid at $\log g = 4.5$, and stars with $\log g < 3.5$ at $\log g = 2.5$, matching the dwarf–giant scheme of §4.3.2. Stars lacking a Gaia $\log g$ value are defaulted to the dwarf grid; these constitute a small fraction of the sample at the magnitudes considered here.

For each catalogued star, the color correction $\Delta m_{\text{color}}(T_{\text{eff}})$ is computed by querying the ATLAS9 grid at the star’s T_{eff} and class-appropriate $\log g$, then evaluating Eq. 4.21 as in the class-representative analysis.

The catalogued Gaia G-band magnitude is converted to the system bandpass through a per-star synthetic transformation derived from the same ATLAS9 SED. The Gaia G band is on the Vega magnitude system, defined analogously to m_V (Eq. 4.16) but evaluated through the Gaia G bandpass:

$$m_G = -2.5 \log_{10} \frac{\dot{C}_G^*}{\dot{C}_G^{\text{Vega}}}. \quad (4.28)$$

The system-bandpass magnitude m_{sys}^* from Eq. 4.18 is then

$$m_{\text{sys}}^* = m_G + \Delta m_{G \rightarrow \text{sys}}(T_{\text{eff}}), \quad (4.29)$$

where the per-star transformation is

$$\Delta m_{G \rightarrow \text{sys}}(T_{\text{eff}}) = -2.5 \log_{10} \left(\frac{\dot{C}_{\text{sys}}^* / \dot{C}_{\text{sys}}^{\text{Vega}}}{\dot{C}_G^* / \dot{C}_G^{\text{Vega}}} \right). \quad (4.30)$$

The transformation is therefore the analog of the color correction Δm_{color} derived in §4.2.3, but with Gaia G playing the role of reference bandpass instead of Bessell V. This ensures that the catalogued Gaia magnitude and the system limiting magnitude are compared in the same photometric system. The detectability of each star then follows from Eq. 4.26, evaluated at the airmass corresponding to the field’s elevation: a star is detectable when $m_{\text{sys}}^* < m_{\text{lim}}(T_{\text{eff}}, X)$.

The detection criterion of Eq. 4.26 partitions the catalogued Gaia sample into detectable and undetectable subsets at any chosen observing geometry. To visualize how this partition depends on stellar properties, each star is placed in the (T_{eff}, M_G) plane, with the absolute G magnitude computed from the apparent G magnitude and parallax via the standard distance modulus. The plane is then binned, and within each bin the fraction of stars satisfying the detection criterion is computed. The resulting

per-bin detection rate is the natural figure of merit for the detectable population: it shows directly where on the HR diagram the system is sensitive and where it is not.

The detection rate is estimated by kernel-density estimation rather than by binning individual stars. The pooled Gaia DR3 sample is divided into the full population n_{all} and the subset passing the per-star detection criterion n_{det} ; a Gaussian kernel (bandwidth 0.1 in $\log T_{\text{eff}} - M_G$) is fit to each, and the rate in a cell is the density weighted ratio $f_{\text{det}} = (\rho_{\text{det}} n_{\text{det}}) / (\rho_{\text{all}} n_{\text{all}})$, clipped to $[0, 1]$. Cells in which the total stellar density falls below 4% of its peak are left uncolored, as the ratio is statistically unreliable where only a few stars contribute. The density floor suppresses the sparsely populated hot ($T_{\text{eff}} > 9000$ K) region, where the kernel estimate is unreliable; these stars are therefore shown individually, with detected and undetected cases marked separately. The smoothed region traces the main sequence and giant branch, where the rate is well sampled.

4.4 LEO Object Detectability

The detectability framework developed in §4.3 for reference stars carries over to LEO target detection with two methodological adjustments. First, the SNR formalism is specialized to the satellite-rate stack S_{sat} rather than the sidereal stack, since the LEO target is co-aligned across the data cube by the synthetic-tracking shift. Second, the LEO target’s spectral energy distribution is fundamentally different from a stellar SED: it is the solar spectrum filtered by the spectral reflectance of the satellite’s surface materials and modulated by the Sun-satellite-observer geometry. The reflected-sunlight SED introduces material-dependent color corrections and a new set of geometric parameters — slant range along the line of sight and phase angle — that do not appear in the stellar case.

This subsection is organized in four parts. The SNR formalism for the satellite-rate stack is developed in §4.4.1, specializing the synthetic-tracking SNR (Eq. 4.8) to the satellite-as-point-source geometry¹. The reflected-sunlight SED model is con-

¹For nearly all LEO targets observed by the system used in this work, the satellite is unresolved at the system’s PSF scale and is treated as a point source in the photometric integral. Large nearby objects such as the ISS are excluded, as their angular extent requires extended-source photometry.

structured in §4.4.2, combining the solar SED with the spectral reflectances of representative spacecraft surface materials. The geometric dependencies on slant range and phase angle are also treated in §4.4.2, where the satellite's intrinsic properties (cross-section, material) are converted to an observed magnitude that depends on the satellite's position and orientation relative to the Sun and observer. The material-resolved detectability (§4.4.3) combines the limiting magnitude with the apparent-magnitude model to derive the smallest detectable satellite cross-section for each representative material and orbit altitude. Finally, §4.4.4 develops the parameter-sensitivity framework for the minimum detectable LEO cross-section, including the single-parameter dependencies on read noise, collecting area, and throughput, the material-dependent wavelength-resolved sensitivity, and the operational sensitivity to the geometric parameters of slant range and phase angle.

4.4.1 SNR formalism for LEO target detection

The LEO target is recovered through synthetic tracking (§4.2.2) in the satellite-rate stack S_{sat} . The satellite-rate-stack SNR is therefore identical in form to the sidereal-stack SNR for reference stars (Eq. 4.25), with the source counts now from the satellite rather than a star.

The detected counts C_{sat} follow from the photon-counting integral (Eq. 4.23) evaluated for the satellite's reflected-sunlight SED, developed below, integrated over t_{exp} . The limiting magnitude for a LEO target of given material composition observed at airmass X is

$$m_{\text{lim}}^{\text{LEO}}(\text{material}, X) = m_{\text{lim,Vega}}^{(t_{\text{exp}})} - \Delta m_{\text{color}}^{\text{LEO}}(\text{material}) - k_{\text{eff}}^{\text{LEO}}(\text{material}) X, \quad (4.31)$$

where $\Delta m_{\text{color}}^{\text{LEO}}(\text{material})$ and $k_{\text{eff}}^{\text{LEO}}(\text{material})$ are the color correction and photon-weighted extinction coefficient for the satellite's reflected-sunlight SED, computed by the same machinery as their stellar counterparts (Eqs. 4.21 and 4.27), and over the same band-pass.

4.4.2 Reflected-sunlight SED model

The full SED model combines four components: the solar SED, the spectral reflectance of the satellite's illuminated surface, the geometric flux dilution from slant range, and

the phase-angle dependence of the reflection.

The illuminating spectrum is taken from the CALSPEC solar-analog reference spectrum [Castelli and Kurucz, 2003]. The spectrum is calibrated to the solar irradiance at 1 AU, so the photon flux at Earth from a reflecting satellite of effective cross-section A_{eff} at a slant range r along the line of sight under phase function $\phi(\alpha)$ at phase angle α is

$$F_{\lambda}^{\text{LEO}}(\lambda) = F_{\lambda}^{\odot}(\lambda) R(\lambda) \frac{A_{\text{eff}}}{\pi r^2} \phi(\alpha), \quad (4.32)$$

where $F_{\lambda}^{\odot}(\lambda)$ is the solar SED in $\text{W m}^{-2} \text{nm}^{-1}$ at 1 AU and $R(\lambda)$ is the dimensionless spectral reflectance of the satellite's illuminated surface. The effective cross-section A_{eff} encapsulates both the satellite's projected geometric area in the direction of the Sun and the average reflectance distribution of the surface.

The spectral reflectance $R(\lambda)$ depends on the dominant material exposed to sunlight. Four representative materials are considered: multi-layer insulation (MLI) [Rodriguez et al., 2007], Z93 white paint [Wilkes et al., 1993], fused-silica coverglass [Henninger, 1971], and protected aluminum [Layertec GmbH, 2025]. Figure 4.7 shows the spectral reflectance curves of these materials over the system bandpass.

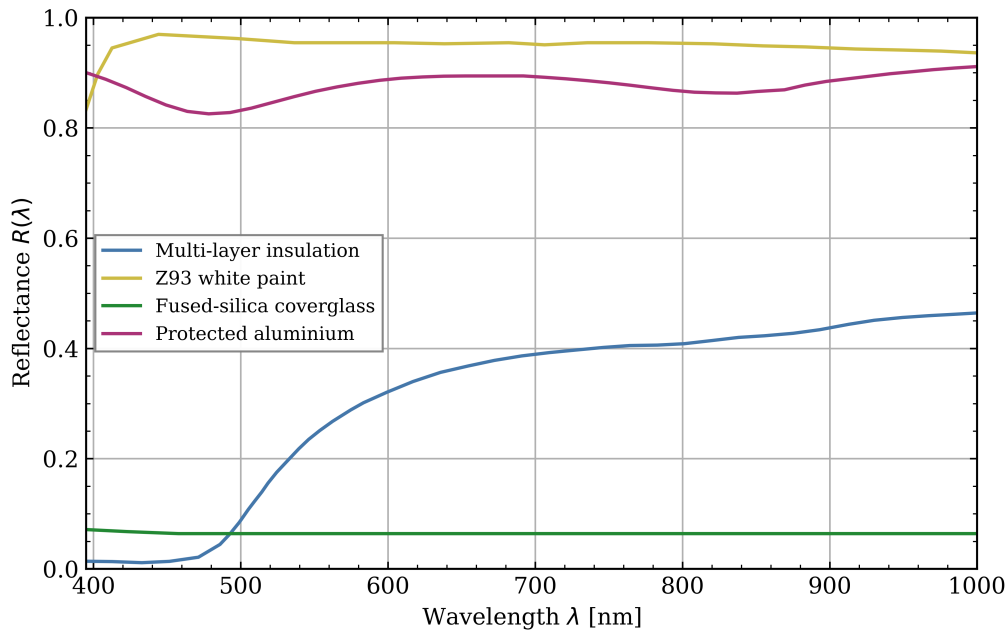


Figure 4.7: Material reflectance curves. The MLI curve is digitized from the space-facing copper-Kapton sample of Rodriguez et al. [2007]; the Z93 from the NASA LDEF Thermal Control Experiment; the fused-silica coverglass from Henninger [1971]; and the protected aluminum from Layertec GmbH [2025].

The choice of these four materials is motivated by their prevalence in the LEO satellite population. Multi-layer insulation is the dominant thermal-control surface on most modern operational satellites and typically wraps the primary structure of communications, observation, and crewed spacecraft [Tachikawa et al., 2022]. Solar-cell coverglass is visible whenever solar arrays are deployed, which includes essentially all powered satellites [NASA Small Spacecraft Systems Virtual Institute, 2024]. White paint dominates the exposed surfaces of payload housings, instrument shrouds, and scientific-satellite optics, where thermal radiation rather than insulation is the design priority [Gilmore, 2002]. Protected aluminum is present as exposed structural elements on most spacecraft and as the dominant surface of older satellites and rocket bodies where MLI was not used or has degraded [Tribble, 2003]. The four materials together span both the dominant operational coverage of the LEO population and a wide range of spectral reflectance shapes — from the broadband-flat response of aluminum and white paint to the strongly red-sloped response of MLI and the bandgap-structured response of solar cells. This spectral diversity is essential for the color-correction analysis (§4.4.4), which depends on the convolution of the reflectance with the system bandpass.

Slant-range and phase-angle dependence

This subsection develops the geometric model that converts a satellite’s intrinsic properties (cross-section, material) and orbit (altitude, elevation) into a predicted apparent magnitude. The slant range r is the line-of-sight distance from the observer to the satellite. For a satellite at altitude h observed at elevation angle ϵ above the horizon, geometry gives

$$r(\epsilon, h) = -R_{\oplus} \sin \epsilon + \sqrt{R_{\oplus}^2 \sin^2 \epsilon + h^2 + 2R_{\oplus}h}, \quad (4.33)$$

where $R_{\oplus} = 6378$ km is the Earth’s mean radius. At zenith ($\epsilon = 90^\circ$), $r = h$; at low elevation, the slant range can exceed the altitude by a factor of two or more. For representative LEO altitudes considered in this work ($h \sim 400\text{--}1200$ km), the slant range varies from h at zenith to $\sim 2\text{--}3h$ at $\epsilon = 30^\circ$.

The apparent flux at Earth scales as r^{-2} from the inverse-square law applied to the

diffuse reflection, giving a magnitude correction of

$$\Delta m_{\text{range}}(r) = +5 \log_{10} \left(\frac{r}{r_0} \right), \quad (4.34)$$

relative to a reference range r_0 ; this work adopts $r_0 = 1000$ km as a fiducial LEO range, so that $m_{\text{app,ref}}$ corresponds to the satellite's apparent magnitude at a slant range of 1000 km and at opposition.

The phase angle α is the angle between the Sun-to-satellite vector and the satellite-to-observer vector. $\alpha = 0^\circ$ corresponds to opposition (the satellite is fully illuminated as seen from Earth), and $\alpha = 180^\circ$ to conjunction (back-illumination). For ground-based observations of LEO satellites in twilight conditions, α typically lies in the range 30° – 120° . The reference geometry $\alpha = 90^\circ$ adopted throughout this work corresponds approximately to the Sun at solar elevation $\epsilon_\odot \approx -18^\circ$, the threshold of astronomical twilight, for a satellite at zenith. At this geometry the sky has darkened to its astronomical-night value of $\mu_{\text{sky}} = 20.5$ mag/arcsec² assumed in the noise budget.

For a Lambertian (diffuse) reflector, the phase function is

$$\phi_{\text{Lamb}}(\alpha) = \frac{\sin \alpha + (\pi - \alpha) \cos \alpha}{\pi}, \quad (4.35)$$

which equals 1 at opposition and falls to zero at conjunction, with $\phi(90^\circ) = 1/\pi$ at quadrature. Real spacecraft surfaces deviate from pure Lambertian behavior depending on the specularity of the reflectance, but the Lambertian approximation is adequate for the diffuse-reflectance regime relevant for typical LEO observations [Linares et al., 2014].

Combining the slant-range and phase-angle dependencies, the apparent magnitude of a satellite of effective cross-section A_{eff} , material reflectance $R(\lambda)$, observed at elevation ϵ and phase angle α , is

$$m_{\text{app}} = m_{\text{app,ref}}(\text{material}, A_{\text{eff}}) + 5 \log_{10} \left(\frac{r(\epsilon, h)}{r_0} \right) - 2.5 \log_{10} \phi(\alpha), \quad (4.36)$$

where $m_{\text{app,ref}}$ is the satellite's apparent magnitude at reference range r_0 and unit phase function. The detectability condition is then

$$m_{\text{app}} < m_{\text{lim}}^{\text{LEO}}(\text{material}, X(\epsilon)), \quad (4.37)$$

where $X(\epsilon)$ is the airmass at the satellite's elevation. This is the operation criterion used in §4.4.3 to determine the smallest detectable satellite as a function of orbit parameters and material composition.

4.4.3 Material-resolved detectability

The detectability condition of Eq. 4.37 provides a yes/no answer for a satellite of known cross-section, but the operational question is the inverse: for a given material, altitude, and viewing geometry, what is the smallest satellite the system can detect? This subsection inverts the detectability condition to derive the minimum detectable effective cross-section $A_{\text{eff,min}}$, the central figure of merit for LEO target detection in this work.

The detected count rate from a satellite of effective cross-section A_{eff} follows from the photon-counting integral of Eq. 4.23 with the reflected-sunlight SED of Eq. 4.32 substituted for the source. The geometric and phase factors are wavelength-independent and factor out of the integral. Introducing the reference count rate

$$\dot{C}_{\text{LEO,ref}}(\text{material}) = \frac{A_{\text{tel}}}{hc} \int_{400 \text{ nm}}^{1000 \text{ nm}} \lambda F_{\lambda}^{\odot}(\lambda) \frac{R(\lambda)}{\pi} \mathcal{T}_{\text{sys}}(\lambda) d\lambda, \quad (4.38)$$

which is the count rate produced at the detector by a satellite of unit effective cross-section at unit slant range, viewed at opposition through no atmosphere, the detected counts integrated over t_{exp} are

$$C^{\text{LEO}} = \dot{C}_{\text{LEO,ref}}(\text{material}) \frac{A_{\text{eff}}}{r^2} \phi(\alpha) T_{\text{atm}}(X, \text{material}) t_{\text{exp}}, \quad (4.39)$$

where $T_{\text{atm}}(X, \text{material}) = 10^{-k_{\text{eff}}^{\text{LEO}}(\text{material}) X/2.5}$ is the photon-weighted atmospheric transmission for the material's reflected SED at airmass X , and $k_{\text{eff}}^{\text{LEO}}$ is defined by Eq. 4.27 with the LEO SED in place of the stellar SED.

Setting Eq. 4.39 equal to the threshold counts C obtained from the SNR inversion of Eq. 4.7, with the noise variance σ_{bg}^2 specialized to the satellite-rate stack of §4.4.1, and solving for A_{eff} gives the minimum detectable effective cross-section

$$A_{\text{eff,min}}(\text{material}, h, \epsilon, \alpha) = \frac{C r^2(\epsilon, h)}{\dot{C}_{\text{LEO,ref}}(\text{material}) \phi(\alpha) T_{\text{atm}}(X, \text{material}) t_{\text{exp}}}. \quad (4.40)$$

This is the operational definition used throughout the results. The reference count rate $\dot{C}_{\text{LEO,ref}}$ and the photon-weighted extinction $k_{\text{eff}}^{\text{LEO}}$ depend only on the material's spectral reflectance and are therefore computed once per material and reused across all geometries.

For interpretation, A_{eff} is more naturally expressed as an equivalent diameter d , defined as the diameter of a circular Lambertian disc with the same effective cross-

section,

$$d = 2\sqrt{\frac{A_{\text{eff}}}{\pi}}. \quad (4.41)$$

The conversion is direct and provides a more intuitive size scale for satellites, which would otherwise need to be inferred from a dimensionless area. Both quantities are reported in the results.

The minimum detectable cross-section is presented in §5.2.2 across the altitude range $h = 400\text{--}1200$ km, elevations $\epsilon = 20^\circ\text{--}90^\circ$, and the four representative materials of §4.4.2. The phase angle is fixed at $\alpha = 90^\circ$ (quadrature) at first as a representative twilight-pass geometry; the phase-angle dependence is presented separately to isolate the geometric contribution from the orbit and elevation contributions.

4.4.4 Parameter-sensitivity framework for LEO targets

The minimum detectable cross section $A_{\text{eff,min}}$ of Eq. 4.40 depends on the system's read noise, collecting area, throughput, and photometric aperture, on the wavelength shape of the system throughput interacting with each material's reflected SED, and on the geometric parameters slant range and phase angle. The sensitivities to each of these quantities are derived below.

System-parameter sensitivities. The partial derivatives of $A_{\text{eff,min}}$ with respect to the system parameters are evaluated numerically by finite difference around the nominal operating point. Closed-form analytical expressions are not used because the noise budget retains contributions from sky background and source shot noise alongside read noise (§4.2.1). The resulting per-parameter sensitivities are reported in §5.2.3.

In the limiting case where the read-noise term dominates the SNR denominator entirely, the analytical scaling can be recovered. Eq. 4.8 reduces to $C_\star = \text{SNR}_{\text{th}} \sqrt{N n_{\text{pix}}} \sigma_{\text{rd}}$, and the chain rule applied to Eq.4.40 yields the pure read-noise limit sensitivities

$$\left(\frac{\partial A_{\text{eff,min}}}{\partial \sigma_{\text{rd}}} \right)_{\text{RN}} = \frac{A_{\text{eff,min}}}{\sigma_{\text{rd}}}, \quad \left(\frac{\partial A_{\text{eff,min}}}{\partial A_{\text{tel}}} \right)_{\text{RN}} = -\frac{A_{\text{eff,min}}}{A_{\text{tel}}}, \quad (4.42)$$

$$\left(\frac{\partial A_{\text{eff,min}}}{\partial \mathcal{T}_{\text{sys}}} \right)_{\text{RN}} = -\frac{A_{\text{eff,min}}}{\mathcal{T}_{\text{sys}}}, \quad \left(\frac{\partial A_{\text{eff,min}}}{\partial n_{\text{pix}}} \right)_{\text{RN}} = \frac{A_{\text{eff,min}}}{2 n_{\text{pix}}}. \quad (4.43)$$

The subscript “RN” denotes that these expressions only apply in the pure read-noise limit. The analytical forms are retained as upper bounds and as a useful guide to

the parameter-dependence of $A_{\text{eff,min}}$: the actual derivatives share the same sign and approximate functional form, but their absolute values are reduced. The number of frames N does not appear as a free parameter because it is constrained by the synthetic-tracking optimum $\tau_{\text{opt}} = t_{\text{exp}}/N$ derived in §4.2.1.

Wavelength-resolved sensitivity. For upgrades that change the shape of the system throughput $\mathcal{T}_{\text{sys}}(\lambda)$ rather than its amplitude alone, the scalar throughput sensitivity is insufficient. The throughput at each wavelength contributes differently to a satellite's detected counts depending on how much reflected flux the satellite delivers there, so a shape change at one wavelength can have a different impact than a shape change of equal magnitude elsewhere. The relevant sensitivity is therefore a function of wavelength rather than a scalar.

Differentiating the photon-counting integral (Eq. 4.38) at fixed reflected SED gives, in the pure read-noise limit,

$$\left(\frac{\partial A_{\text{eff,min}}}{\partial \mathcal{T}_{\text{sys}}(\lambda)} \right)_{\text{RN}} = -A_{\text{eff,min}} \frac{\lambda F_{\lambda}^{\odot}(\lambda) R(\lambda) T_{\text{atm}}(\lambda, \epsilon)}{\int_{400 \text{ nm}}^{1000 \text{ nm}} \lambda F_{\lambda}^{\odot}(\lambda) R(\lambda) \mathcal{T}_{\text{sys}}(\lambda) T_{\text{atm}}(\lambda, \epsilon) d\lambda}, \quad (4.44)$$

which depends on wavelength through the product $F_{\lambda}^{\odot}(\lambda) R(\lambda) T_{\text{atm}}(\lambda, \epsilon)$. The geometric and phase factors of Eq. 4.32 cancel between the numerator and denominator while the material reflectance $R(\lambda)$ makes this sensitivity material-dependent. The same hardware upgrade therefore yields material-dependent improvements in $A_{\text{eff,min}}$, and the per-material sensitivity functions are presented in §5.2.3.

Geometric-parameter sensitivities The slant range r and phase angle ϕ parameters are not under the observer's direct control in the same way as system parameters, but the resulting sensitivities quantify the operational impact of choosing one observing geometry over another.

The slant-range derivative follows directly from $A_{\text{eff,min}} \propto r^2$ in Eq. 4.40:

$$\frac{\partial A_{\text{eff,min}}}{\partial r} = \frac{2 A_{\text{eff,min}}}{r}. \quad (4.45)$$

A 10% increase in slant range yields a 21% increase in $A_{\text{eff,min}}$, or a 10% increase in equivalent diameter. The slant range itself depends on altitude h and elevation ϵ through Eq. 4.33, so this derivative implicitly captures the sensitivity to both quantities.

The phase-angle derivative is obtained by differentiating the Lambertian phase function of Eq. 4.35:

$$\frac{d\phi}{d\alpha} = -\frac{(\pi - \alpha) \sin \alpha}{\pi}. \quad (4.46)$$

Combined with $A_{\text{eff,min}} \propto 1/\phi(\alpha)$ from Eq. 4.40, the chain rule gives

$$\frac{\partial A_{\text{eff,min}}}{\partial \alpha} = A_{\text{eff,min}} \frac{(\pi - \alpha) \sin \alpha}{\pi \phi(\alpha)}. \quad (4.47)$$

The sensitivity vanishes near opposition ($\alpha \rightarrow 0^\circ$), where the satellite is fully illuminated and small angular changes produce no first-order brightness variation, and diverges near conjunction ($\alpha \rightarrow 180^\circ$) as the reflected flux approaches zero. For operational planning, the geometric sensitivities formalize the qualitative expectation that LEO targets are easier to detect (smaller $A_{\text{eff,min}}$) when observed close to zenith and away from conjunction. The full geometric dependence is reported in §5.2.3.

Combined-parameter upgrade paths. A realistic hardware upgrade typically changes multiple system parameters at once. A detector swap, for example, generally changes both the read noise σ_{rd} and the wavelength-dependent quantum efficiency entering $\mathcal{T}_{\text{sys}}(\lambda)$. To quantify the impact of such an upgrade, a scalar u parameterizes the upgrade path, with each affected parameter changing along the path as a known function $p_i(u)$. The total change in $A_{\text{eff,min}}$ along the path is the directional derivative

$$\frac{dA_{\text{eff,min}}}{du} = \sum_i \frac{\partial A_{\text{eff,min}}}{\partial p_i} \frac{dp_i}{du}, \quad (4.48)$$

where the sum runs over all parameters p_i that change with u . The single-parameter derivatives $\partial A_{\text{eff,min}}/\partial p_i$ are evaluated as above; the coupling functions dp_i/du are set by the physics of the specific upgrade rather than by the SNR formalism.

For upgrades that change the shape of $\mathcal{T}_{\text{sys}}(\lambda)$ in addition to scalar parameters, the scalar $\partial A_{\text{eff,min}}/\partial \mathcal{T}_{\text{sys}}$ in Eq. 4.48 is replaced by the wavelength integral

$$\left(\frac{dA_{\text{eff,min}}}{du} \right)_{\mathcal{T}} = \int_0^\infty \frac{\partial A_{\text{eff,min}}}{\partial \mathcal{T}_{\text{sys}}(\lambda)} \frac{d\mathcal{T}_{\text{sys}}(\lambda)}{du} d\lambda, \quad (4.49)$$

with the wavelength-resolved sensitivity given by Eq. 4.44. In practice, for upgrades with large changes in QE shape, the integral approach linearizes around the nominal throughput and may not capture the full effect; the end-to-end response is then better evaluated by recomputing $A_{\text{eff,min}}$ directly with the upgraded throughput substituted into the noise budget. The specific upgrade scenarios considered are analyzed in §5.2.3.

Results

The detectability framework developed in §4 is applied here to quantify the system’s performance against the two target classes of interest: astrometric reference stars (§5.1) and LEO targets (§5.2). Throughout, the calibration outputs derived in §4.1 — the empirically scaled system throughput $\mathcal{T}_{\text{sys}}(\lambda)$, the atmospheric transmission grid $T_{\text{atm}}(\lambda, \epsilon)$, and the photometric zero point $ZP = 20.34 \pm 0.22$ mag — serve as the calibrated foundation for the SNR and detectability calculations that follow. The synthetic-tracking observing strategy and its resulting limiting magnitudes are validated against the reference-star formalism in §5.1; the reference-star detectability across a representative Gaia DR3 catalog is presented in §5.1; and the LEO detectability is then computed for the four representative spacecraft materials in §5.2, with a parameter-sensitivity analysis along plausible upgrade paths. Throughout, the elevation $\epsilon = 30^\circ$ is identified as an operating point that balances atmospheric extinction against synthetic-tracking integration time, and is used as the reference geometry for inter-comparison plots.

5.1 Synthetic tracking and limiting magnitude validation

With the calibration outputs of §4.1, the synthetic-tracking framework of §4.2.1 can be evaluated quantitatively. This section reports the synthetic-tracking observation parameters across the altitude–elevation grid of interest, validates the resulting limiting-magnitude calculation against the alternative sidereal single-exposure strategy, and presents the spectral-type dependence of the limiting magnitude that anchors the reference-star analysis.

5.1.1 Synthetic-tracking observation parameters

The synthetic-tracking strategy is fully determined at each altitude–elevation point by four quantities: the satellite’s angular velocity across the focal plane $\dot{\theta}$, the optimal sub-exposure duration τ_{opt} , the total integration time t_{exp} set by the transit across the detector long axis, and the resulting number of stacked frames $N = t_{\text{exp}}/\tau_{\text{opt}}$. These four quantities are tabulated in Table 5.1 for the satellite altitude range $h = 400\text{--}1200$ km at elevations spanning $\epsilon = 20^\circ\text{--}90^\circ$.

The competing dependencies of t_{exp} and atmospheric extinction on elevation produce a shallow optimum in the limiting magnitude. At low elevation, the long slant range gives an extended transit time t_{exp} and therefore a large N , but the additional atmospheric extinction shifts the limiting magnitude brighter. At high elevations, the transit is short and the sky is at its darkest, but N is correspondingly small. The optimum sits near $\epsilon \approx 30^\circ$, which is used as the reference geometry for inter-comparison plots throughout the remainder of this work; the full elevation dependence is tabulated, and the choice of reference elevation is robust to modest changes in the read-noise and sky-brightness assumptions.

The elevation dependence of the noise budget is driven primarily by the sky background. The per-pixel sky count rate $\dot{C}_{\text{sky}}(X)$, derived from the system zero point and the assumed dark-sky surface brightness of $20.5 \text{ mag/arcsec}^2$, scales linearly with airmass, growing from $\sim 15 \text{ e}^-/\text{pix/s}$ at zenith to $\sim 75 \text{ e}^-/\text{pix/s}$ at $\epsilon = 20^\circ$ (Figure 5.1). Combined with the per-frame read noise, this sets the variance in the photometric aperture that enters the limiting-magnitude calculation.

Angular velocity $\dot{\theta}$ [arcsec/s]							Optimal sub-exposure τ_{opt} [ms]						
h [km]	Elevation ϵ						h [km]	Elevation ϵ					
	20°	30°	45°	60°	75°	90°		20°	30°	45°	60°	75°	90°
400	1607	2139	2876	3458	3828	3954	400	52.2	54.7	56.1	56.6	56.8	56.9
500	1316	1726	2299	2752	3041	3140	500	57.6	60.9	62.7	63.4	63.7	63.8
600	1120	1450	1913	2282	2517	2598	600	62.5	66.4	68.7	69.7	70.0	70.2
800	869	1102	1431	1694	1863	1921	800	70.9	76.2	79.5	80.8	81.4	81.6
1000	715	891	1141	1342	1472	1516	1000	78.2	84.7	89.0	90.8	91.6	91.8
1200	609	748	947	1107	1211	1247	1200	84.7	92.4	97.7	100.0	101.0	101.3

(a)

Total integration time t_{exp} [s]							Number of stacked frames N						
h [km]	Elevation ϵ						h [km]	Elevation ϵ					
	20°	30°	45°	60°	75°	90°		20°	30°	45°	60°	75°	90°
400	1.5	1.1	0.8	0.7	0.6	0.6	400	27	19	14	11	10	10
500	1.8	1.4	1.0	0.8	0.8	0.7	500	30	22	16	13	12	11
600	2.1	1.6	1.2	1.0	0.9	0.9	600	33	24	17	14	13	12
800	2.7	2.1	1.6	1.4	1.3	1.2	800	37	27	20	17	15	14
1000	3.3	2.6	2.0	1.7	1.6	1.5	1000	41	30	22	19	17	16
1200	3.8	3.1	2.5	2.1	1.9	1.9	1200	45	33	25	21	19	18

(c)

(d)

Table 5.1: Synthetic-tracking observation parameters as a function of LEO satellite altitude h and elevation angle ϵ , for (a) angular velocity, (b) optimal sub-exposure duration, (c) total integration time, and (d) number of stacked frames. Computed for the 8'' SCT system with the ZWO ASI432MM detector (1608×1104 pixels at $1.45''/\text{pixel}$, $\text{FOV } 38.9' \times 26.7'$, with t_{exp} defined as the transit time across the $38.9'$ long axis). The field is rotated to align the detector long axis with the satellite's apparent motion. Sky brightness scales linearly with airmass. The bold values mark the reference geometry ($h = 500$ km, $\epsilon = 30^\circ$).

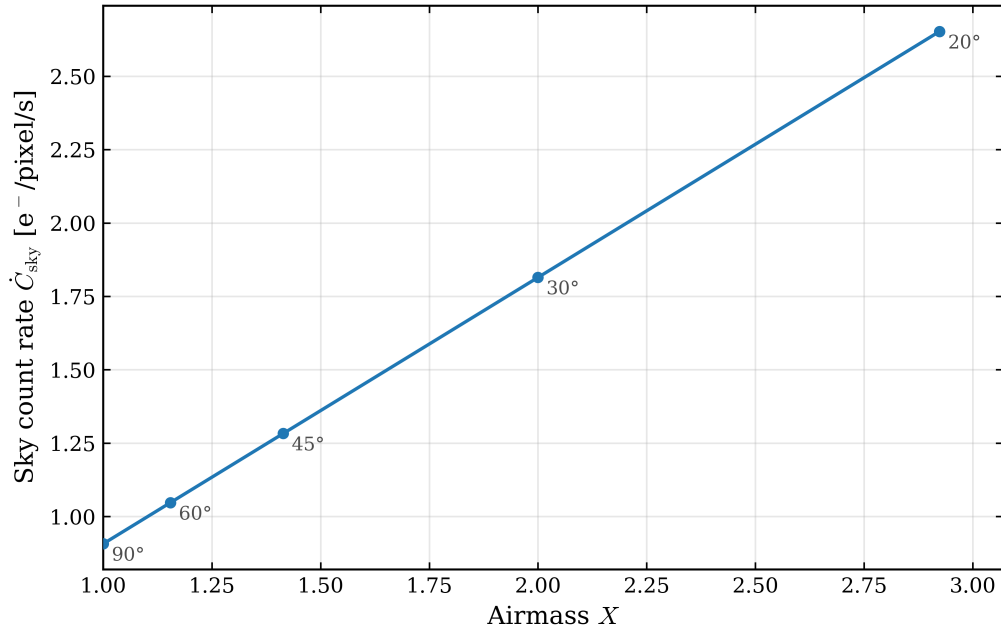


Figure 5.1: Per-pixel sky count rate $\dot{C}_{sky}$ as a function of airmass, derived from the system zero point and an assumed dark-sky surface brightness of 20.5 mag/arcsec².

5.1.2 Per-class limiting magnitudes

The stacked limiting magnitude for the representative spectral classes of §4.3.2 is computed by combining the noise budget at each elevation with the per-class color correction $\Delta m_{color}(T_{eff})$ and photon-weighted extinction $k_{eff}(T_{eff})$, tabulated in Table 5.2. The resulting limiting magnitudes are shown in Figure 5.2 for both main-sequence dwarfs (B5 V through M2 V) and red giants (G8 III, K2 III, M0 III).

Type	T_{eff} [K]	$\log g$	$\Delta m_{G \rightarrow V}$	Δm_{color}	$\Delta m_{\text{sys} \rightarrow G}$	k_{eff}
B5 V	15700	4.5	+0.017	+0.007	+0.024	0.145
A0 V	9700	4.5	-0.006	+0.006	-0.001	0.141
F5 V	6510	4.5	+0.094	-0.098	-0.004	0.128
G2 V	5780	4.5	+0.141	-0.147	-0.006	0.124
K5 V	4400	4.5	+0.395	-0.390	+0.005	0.117
M2 V	3550	4.5	+0.863	-0.815	+0.048	0.118
G8 III	4880	2.5	+0.236	-0.240	-0.004	0.119
K2 III	4500	2.5	+0.319	-0.319	+0.001	0.117
M0 III	3690	2.5	+0.761	-0.722	+0.039	0.117

Table 5.2: Per-class color corrections and photon-weighted atmospheric extinction. All Δm are above-atmosphere, computed from model SEDs integrated through the respective bandpasses and anchored to Vega in V. k_{eff} is the photon-weighted extinction in the system bandpass.

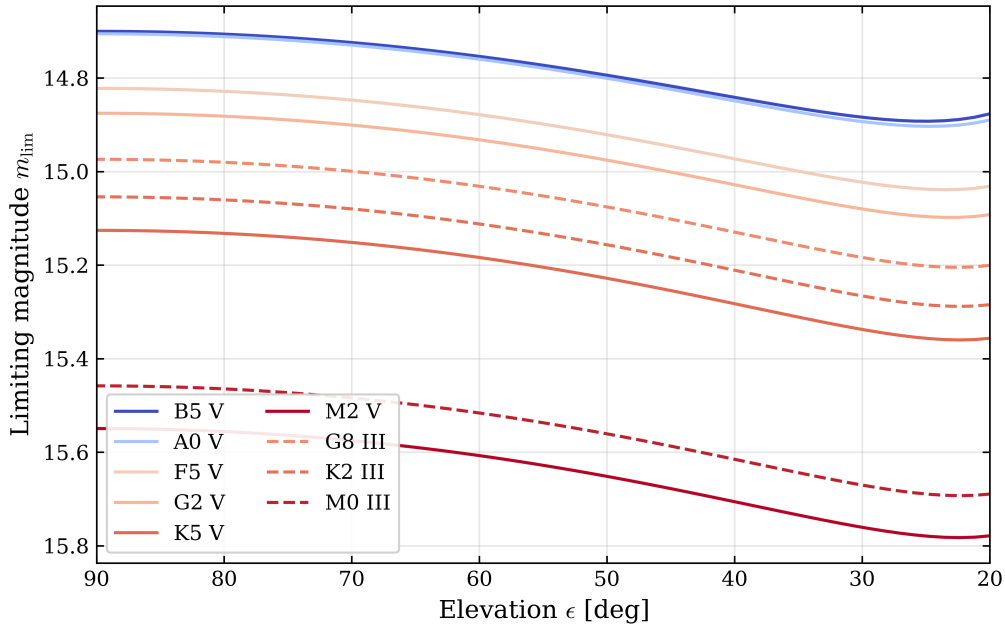


Figure 5.2: Stacked limiting magnitude m_{lim} versus elevation ϵ with $h = 500$ km for main-sequence dwarfs (B5 V–M2 V, solid) and giants (G8 III–M0 III, dashed).

Two trends are visible across the spectral types. First, cooler stars (M2V, K5V) achieve fainter limiting magnitudes than hotter stars (B5V, A0V), reflecting the negative color correction of cool SEDs sampled through the red-extended system band-

pass. Second, the elevation dependence of m_{lim} is mild, with all curves showing a shallow optimum near $\epsilon \approx 30^\circ$ and a peak-to-trough variation of ~ 0.15 mag across the elevation range.

5.1.3 Comparison against single sidereal exposures

Comparing the two observing strategies quantifies the sensitivity cost of stacking and clarifies why it is nonetheless the right choice for this work. Figure 5.3 compares the limiting magnitudes achieved by the two strategies for an M2 V reference star with $h = 500$ km across the elevation range. The single-exposure strategy is deeper at every elevation which is expected in the read-noise-dominated regime.

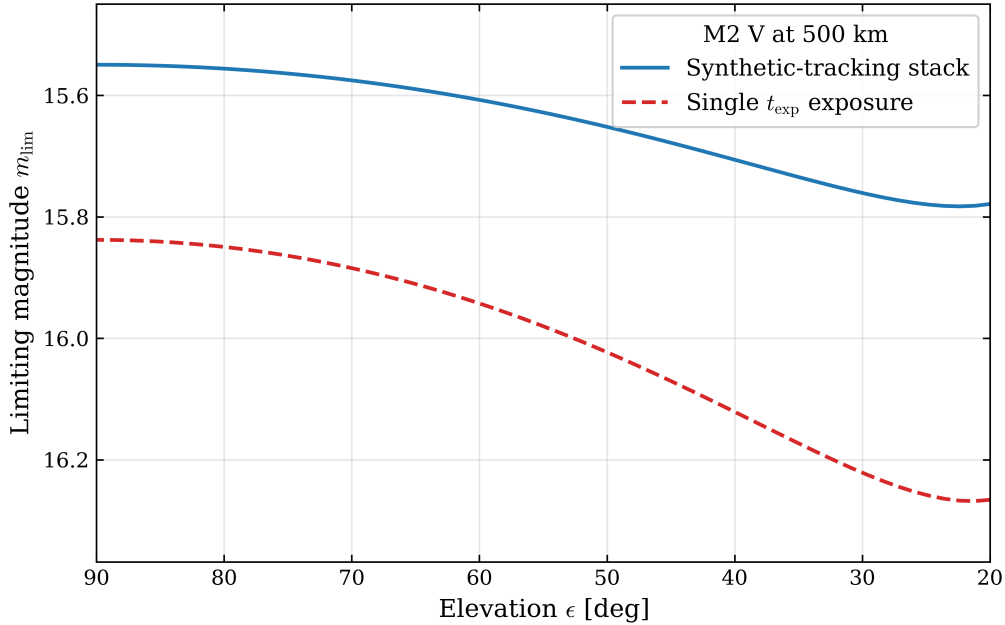


Figure 5.3: Stacked synthetic-tracking limiting magnitude (solid) compared against a single sidereal exposure of the same total duration (dashed), for an M2 V reference star as a function of elevation with $h = 500$ km. The single exposure reaches ~ 0.1 – 0.4 mag deeper at all elevations, the $\sqrt{N}$ read-noise penalty incurred by stacking in the read-noise-dominated regime.

The single exposure is deeper, but it cannot recover the LEO target. The modest ~ 0.1 – 0.4 mag advantage of a single exposure does not justify forfeiting the LEO-target measurement, so reference-star detectability is evaluated using the synthetic-tracking stack throughout (Figure 5.2).

5.1.4 Detectable star population from Gaia DR3

The per-class limiting magnitudes of §5.1.2 characterize the detectability of a representative reference star at each spectral type, but the operationally relevant quantity is the fraction of real stars in the sky that are detectable through the system. This subsection applies the per-star detection criterion of §4.3.3 to a Gaia DR3 sample restricted to the southern sky accessible from the observing site and validates the class-representative tabulation of §4.3.2 against the population-weighted result.

Per-star detection check. For each retained star, the system magnitude is computed from the Gaia G magnitude through the synthetic transformation of Eq. 4.29, with the per-star $\Delta m_{G \rightarrow \text{sys}}$ evaluated at the star's T_{eff} and class-appropriate $\log g$ against the ATLAS9 grid. The detection threshold for the same star is then computed from Eq. 4.26 at the observation airmass, using the per-star photon-weighted extinction coefficient $k_{\text{eff}}(T_{\text{eff}})$ from Eq. 4.27. A star is flagged as detectable when its system magnitude is brighter than its temperature-dependent threshold. The resulting population of detected and rejected stars is shown on the (T_{eff}, M_G) plane in Figure 5.4.

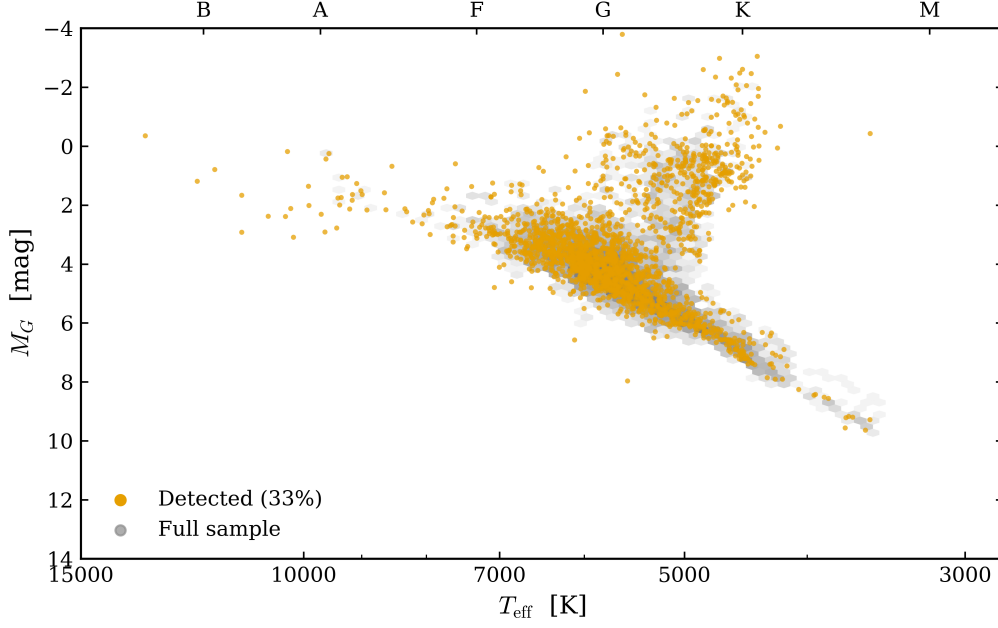


Figure 5.4: (T_{eff}, M_G) diagram of the Gaia DR3 sample pooled across the six representative fields, with the detection criterion evaluated at the reference geometry. Orange points are flagged as detectable; gray points are stars below the per-star limiting magnitude. Absolute G magnitude is computed from the apparent Gaia G magnitude and parallax via the standard distance modulus. Spectral-type letter across the top correspond to the standard T_{eff} ranges.

Field densities and reference-star yield. Table 5.3 summarizes the detectable population in each field at the reference geometry, together with the implied reference-star counts under two distinct field-of-view definitions: the full $38.9' \times 26.7'$ sensor area used during plate-solving calibration exposures, and the narrow $38.9' \times 14.5''$ region-of-interest strip used during synthetic-tracking science exposures (§3.1). The full-FOV counts range from 105 to 1112 stars across the six fields, while the ROI counts range from 1.42 to 15.11. The ROI is the binding case: the synthetic-tracking science exposure must contain at least one reference stars to determine the object’s location at each frame. The plate-solving FOV provides a comfortable margin in all cases.

Field	b [°]	N_{retained}	Stars/arcmin ²	Per FOV	Per ROI
Low-latitude A	−15	2034	0.72	746	10.13
Low-latitude B	−10	3032	1.07	1112	15.11
Mid-latitude A	−42	506	0.18	186	2.52
Mid-latitude B	−52	491	0.17	180	2.45
High-latitude A	−62	343	0.12	126	1.71
High-latitude B	−76	286	0.10	105	1.42

Table 5.3: Gaia DR3 reference-star yield across the six representative fields. N_{retained} counts stars satisfying the $G < 16.5$ magnitude cap, the GSP-Phot stellar-parameter requirements, and the ATLAS9 grid coverage cut on T_{eff} . The “Per FOV” column reports the expected number of reference stars in the full $38.9' \times 26.7'$ plate-solving exposure; the “Per ROI” column reports the same quantity in the synthetic-tracking region of interest ($38.9' \times 14.5''$).

Detectability across the (T_{eff}, M_G) plane. Figure 5.5 shows the detection rate across the (T_{eff}, M_G) plane at the reference geometry, with effective temperature on the horizontal axis and absolute G magnitude M_G on the vertical axis. Regions of the diagram with high detection probability are shaded yellow; regions with low probability are shaded dark blue.

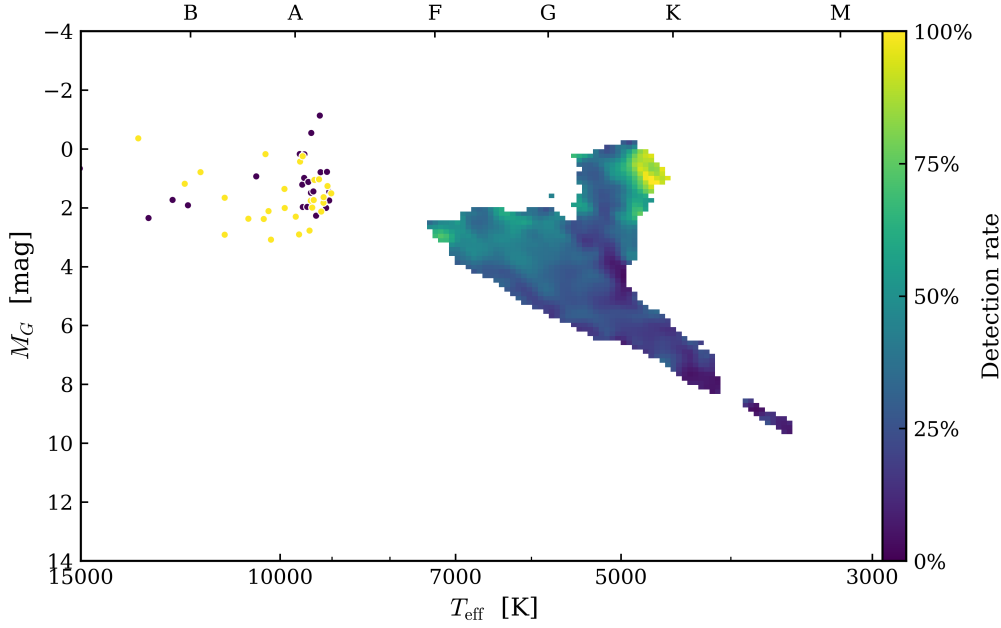


Figure 5.5: Detection rate across the (T_{eff}, M_G) plane at the reference geometry, computed across the Gaia DR3 sample pooled from all six fields. The smoothed map shows the KDE-estimated fraction of stars satisfying the per-star detection criterion of §4.3.3; individual points mark hot stars ($T_{\text{eff}} > 9000$ K) where the kernel estimate is unreliable, with yellow and purple indicating detected and undetected, respectively.

5.2 LEO Target Detectability

The detectability framework developed in §4.4 for LEO targets is applied here to the four representative spacecraft surface materials of §4.4.2: multi-layer insulation (MLI), Z93 white paint, fused-silica coverglass, and protected aluminum. This section first reports the per-material color corrections, extinction coefficients, and limiting magnitudes that follow directly from each material’s reflected-sunlight SED, then converts these into the minimum detectable effective cross-section $A_{\text{eff},\text{min}}$ across the altitude–elevation–phase-angle grid, and finally analyzes the parameter sensitivities along plausible hardware upgrade paths.

5.2.1 Material-resolved limiting magnitude

The color correction $\Delta m_{\text{color}}^{\text{LEO}}$ and photon-weighted extinction coefficient $k_{\text{eff}}^{\text{LEO}}$ for each material are computed by substituting the reflected-sunlight SED of Eq. 4.32 into the

color-correction and extinction integrals (Eqs. 4.21 and 4.27). The resulting values are reported in Table 5.4.

Material	$\Delta m_{\text{color}}^{\text{LEO}}$ [mag]	$k_{\text{eff}}^{\text{LEO}}$ [mag/airmass]	$m_{\text{lim}}^{\text{LEO}}$ [mag]
Multi-layer insulation	−0.421	0.109	14.50
Z93 white paint	−0.152	0.124	14.23
Fused-silica coverglass	−0.157	0.125	14.23
Protected aluminium	−0.167	0.124	14.24

Table 5.4: Per-material color correction $\Delta m_{\text{color}}^{\text{LEO}}$, photon-weighted atmospheric extinction coefficient $k_{\text{eff}}^{\text{LEO}}$, and limiting magnitude $m_{\text{lim}}^{\text{LEO}}$ for the four representative spacecraft surface materials. The limiting magnitude is computed at the reference observing geometry ($h = 500$ km, $\epsilon = 30^\circ$, $\alpha = 90^\circ$) and represents the apparent V magnitude of a satellite of that material at the $\text{SNR} = 5$ detection threshold.

The color corrections span approximately -0.42 to -0.15 mag across the four materials. Three of the materials (Z93 white paint, fused-silica coverglass, and protected aluminum) cluster near $\Delta m_{\text{color}}^{\text{LEO}} \approx -0.16$ mag. This common offset reflects the color correction inherent to a solar-like reflected SED: all three materials have approximately flat spectral reflectance, so the reflected light retains the solar SED shape.

MLI is the exception, with $\Delta m_{\text{color}}^{\text{LEO}} = -0.421$ mag. The additional shift reflects MLI’s red-rising reflectance, which climbs from ~ 0.01 at 400 nm to ~ 0.46 at 1000 nm and concentrates flux preferentially in the NIR where the system bandpass remains sensitive but Bessell V has fallen to zero.

The modest spread of the photon-weighted extinction coefficients reflects the broad system bandpass, which samples a range of extinction values that are similar in magnitude across the visible and NIR. MLI’s NIR-weighted SED samples the lowest-extinction portion of the atmosphere and therefore carries the smallest $k_{\text{eff}}^{\text{LEO}}$; the other three materials, with visible-peaked reflected SEDs, cluster at slightly higher values.

Combining the color correction and extinction coefficient with the Vega-equivalent stacked limiting magnitude from §5.1 yields the per-material limiting magnitude $m_{\text{lim}}^{\text{LEO}}$ at each observing geometry.

5.2.2 Detectable cross-section

The per-material limiting magnitude converts to a minimum detectable effective cross-section $A_{\text{eff,min}}$ through Eq. 4.40, which combines the magnitude threshold with the slant-range and phase-function geometric factors. This subsection reports $A_{\text{eff,min}}$ across the altitude–elevation grid at fixed phase angle ($\alpha = 90^\circ$, quadrature) as a representative twilight-pass geometry, and separately treats the phase-angle dependence at the reference altitude.

Diameter versus apparent magnitude for the four materials. The relationship between satellite size and apparent magnitude is shown in Figure 5.6 for the four representative materials at the reference geometry. Each curve follows the expected $m_V \propto -5 \log_{10}(d)$ relation on the log-diameter axis, with the four curves offset vertically by their material-specific integrated reflectance through the system bandpass. The ordering is as expected from the reflectance integrals using the reflectance curves from Figure 4.7: aluminum and Z93 paint achieve the smallest detectable diameters; MLI is intermediate; fused-silica coverglass is the largest.

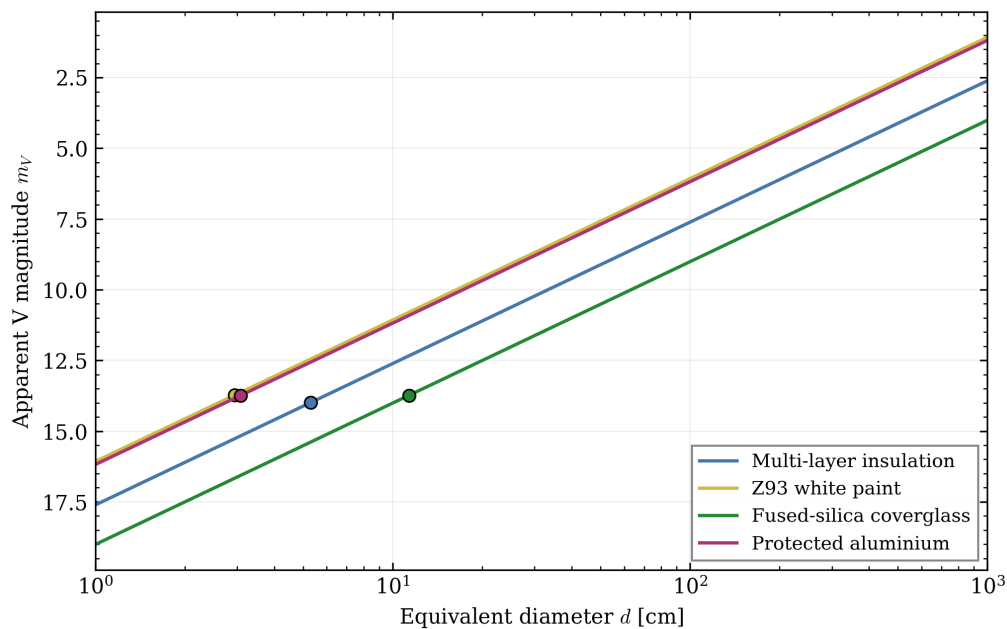


Figure 5.6: Apparent V magnitude as a function of equivalent diameter for the four representative spacecraft surface materials, at the reference geometry. Black-edged markers indicate the minimum detectable diameter at the $\text{SNR} = 5$ threshold for each material.

Elevation dependence of the minimum detectable diameter. The reference geometry of Figure 5.6 captures a single operating point; the full elevation dependence of the minimum detectable diameter $d_{\min}$ is shown in Figure 5.7, at fixed altitude $h = 500$ km and phase angle $\alpha = 90^\circ$.

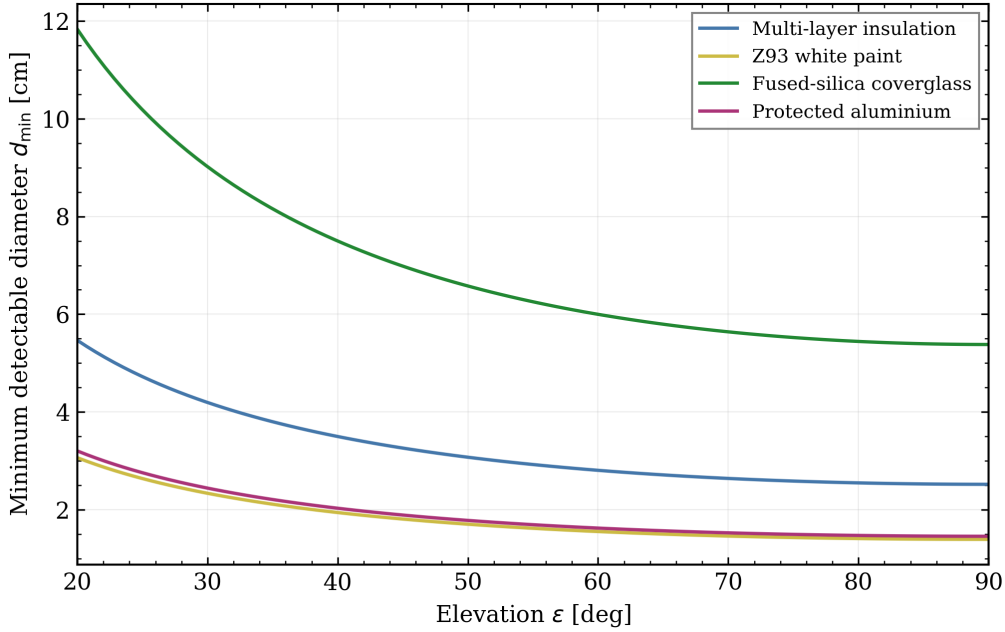


Figure 5.7: Minimum detectable equivalent diameter $d_{\min}$ as a function of observation elevation ϵ for each of the four representative spacecraft surface materials, at fixed altitude $h = 500$ km and phase angle $\alpha = 90^\circ$.

For all four materials, $d_{\min}$ decreases monotonically with increasing elevation and reaches its minimum at zenith. The shape of the curve reflects the competing elevation dependencies in the noise budget: the slant range $r(\epsilon, h)$ scales approximately as $1/\sin \epsilon$ at LEO altitudes, so $A_{\text{eff}, \min} \propto r^2$ grows by roughly a factor of four between zenith and $\epsilon = 30^\circ$; the atmospheric extinction at the same wavelengths grows by ~ 0.3 mag; and the transit time t_{exp} across the detector's long axis lengthens by approximately a factor of two, yielding a $\sqrt{2}$ improvement in stacked SNR. The slant-range and extinction penalties dominate, and the $\sqrt{N}$ gain from a longer integration does not compensate.

This contrasts with the reference-star case of §5.1.4, where the limiting magnitude shows a shallow optimum near $\epsilon \approx 30^\circ$. Reference stars are at effectively infinite distance, so the r^2 term does not enter; only atmospheric extinction and transit time

compete, and they balance to produce the shallow mid-elevation optimum. For LEO targets, the additional r^2 dependence shifts the optimum to zenith. The operational implication is that observations of LEO targets should be conducted at the highest elevation that the pass geometry permits; the reference elevation $\epsilon = 30^\circ$ used in the tabulations of §5.2.1 is retained for comparison with the reference-star analysis, but is not the operationally optimal geometry for LEO detection.

The relative ordering of the four materials is preserved across the elevation range, since the elevation dependence is dominated by geometry and the atmospheric extinction is only weakly material-dependent through $k_{\text{eff}}^{\text{LEO}}(\text{material})$.

Altitude dependence. Figure 5.8 shows the complementary view: d_{min} as a function of satellite altitude at zenith pass ($\epsilon = 90^\circ$, $\alpha = 90^\circ$).

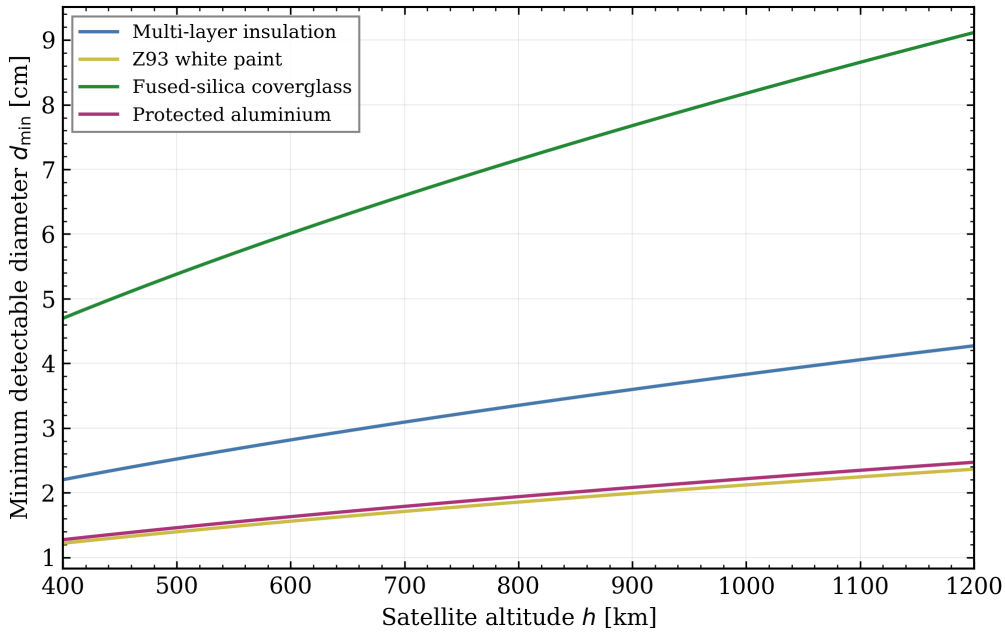


Figure 5.8: Minimum detectable equivalent diameter d_{min} as a function of satellite altitude h for each of the four representative spacecraft surface materials, at zenith pass ($\epsilon = 90^\circ$, $\alpha = 90^\circ$). The approximately linear scaling reflects the $r = h$ geometry at zenith and the $d_{\text{min}} \propto r$ dependence of detectability.

With the satellite directly overhead, the slant range equals the altitude ($r = h$), and the r^2 dependence of $A_{\text{eff,min}}$ translates directly to $d_{\text{min}} \propto h$. Across the $h = 400$ – 1200 km range relevant for the operational LEO population, d_{min} grows by approximately a factor of two for all four materials, consistent with the linear scaling and

modulated only slightly by the increased integration time available for higher-altitude targets. The material ordering is preserved across the altitude range, again confirming that altitude dependence is dominated by geometry rather than by material-specific extinction.

Phase-angle dependence. The third geometric parameter, the Sun-satellite-observer phase angle α enters $A_{\text{eff,min}}$ through the Lambertian phase function $\phi(\alpha)$ of Eq. 4.35. Figure 5.9 shows d_{min} as a function of α at the reference altitude and zenith pass, for each of the four materials. In practice, the accessible range of phase angles for ground-based twilight observations is constrained by the requirement that the satellite be sunlit while the observer's sky is dark, which limits α to approximately 30° – 120° .

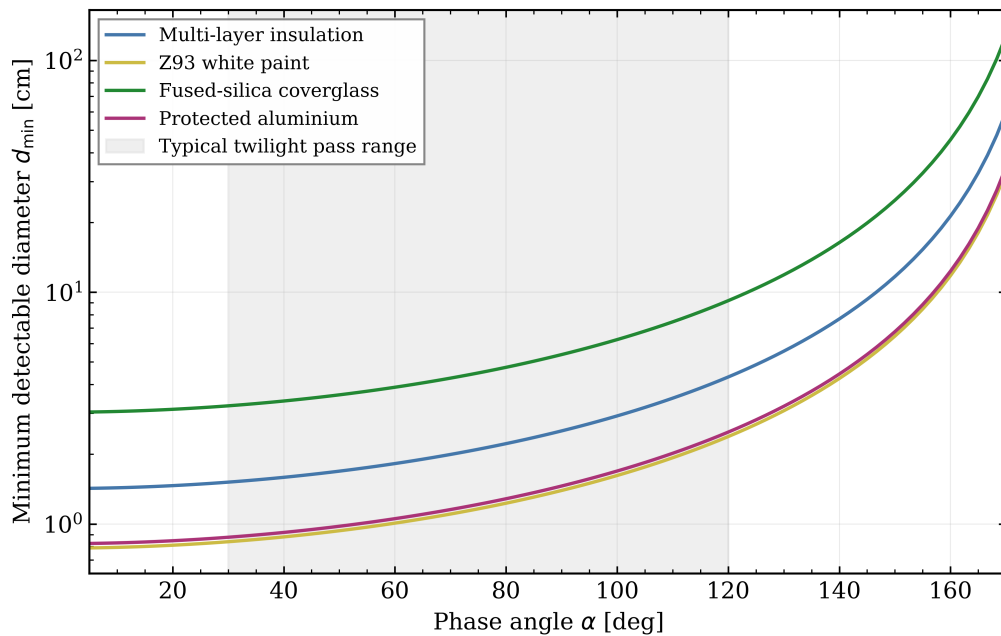


Figure 5.9: Minimum detectable equivalent diameter d_{min} as a function of Sun-satellite-observer phase angle α for each of the four representative spacecraft surface materials, at fixed altitude $h = 500$ km and zenith pass ($\epsilon = 90^\circ$). The shaded band marks the operational phase-angle range for twilight LEO observations ($\alpha \approx 30^\circ$ – 120°).

Summary. Detectability improves monotonically toward zenith, toward lower altitudes, and toward smaller phase angles. The best-case operational geometry within the surveyed range ($h = 400$ km, $\epsilon = 90^\circ$, $\alpha = 30^\circ$) delivers $d_{\text{min}} \approx 0.73$ cm for the Z93 white paint; the worst-case ($h = 1200$ km, $\epsilon = 20^\circ$, $\alpha = 120^\circ$) raises this to 8.10 cm for

the same material, a factor of ~ 11 degradation. The operational implications of this geometric envelope are discussed in Chapter 6.

5.2.3 Parameter-sensitivity for LEO targets

The parameter-sensitivity framework of §4.4.4 is applied here to quantify the response of $A_{\text{eff,min}}$ to realistic hardware upgrades and to the observing geometry. The wavelength-resolved sensitivity is material-dependent, so the response is evaluated separately for each candidate surface material.

Single-parameter sensitivities at the reference operating point. The partial derivatives of $A_{\text{eff,min}}$ with respect to the system parameters at the reference operating point are reported in Table 5.5. The sensitivities are independent of material at this level of analysis because the threshold count rate that defines $A_{\text{eff,min}}$ is set by the noise budget alone; the material only enters through the conversion from cross-section to detected counts, and that conversion does not change with system parameters.

$\partial A_{\text{eff,min}} / \partial \sigma_{\text{rd}}$	$\partial A_{\text{eff,min}} / \partial D$	$\partial A_{\text{eff,min}} / \partial \mathcal{T}_{\text{sys}}$	$\partial A_{\text{eff,min}} / \partial n_{\text{pix}}$
[% / e^-]	[% / %]	[% / %]	[% / %]
+23.48	−2.01	−1.00	+0.49

Table 5.5: Single-parameter sensitivities of the minimum detectable cross-section $A_{\text{eff,min}}$ at the reference operating point ($h = 500$ km, $\epsilon = 30^\circ$, $\alpha = 90^\circ$, $\sigma_{\text{rd}} = 2 e^-$, $D = 20.3$ cm). Read-noise sensitivity is reported per 1-electron change; collecting-area diameter D , throughput $\mathcal{T}_{\text{sys}}$, and photometric pixel count n_{pix} are reported per 1% relative change. Positive values indicate that increasing the parameter degrades detectability (larger $A_{\text{eff,min}}$); negative values indicate improvement.

The dominant parameter is the read noise σ_{rd} : a $1 e^-$ reduction yields a $\sim 24\%$ reduction in $A_{\text{eff,min}}$, corresponding to a $\sim 13\%$ reduction in equivalent diameter. The collecting area sensitivity is also strong: a 1% increase in aperture diameter D reduces $A_{\text{eff,min}}$ by 2%, reflecting the $A \propto 1/D^2$ scaling of the read-noise-dominated limiting magnitude. As an operational benchmark, doubling the aperture from $D = 20.3$ cm to $D = 40.6$ cm reduces $A_{\text{eff,min}}$ by 75%, equivalent to a 50% reduction in equivalent

diameter. The throughput and photometric aperture sensitivities are weaker, with the throughput producing a 1% improvement in $A_{\text{eff,min}}$ per 1% increase in $\mathcal{T}_{\text{sys}}$ and the pixel count contributing $\sim 0.5\%$ per 1% change in n_{pix} .

While Table 5.5 reports the local response at the reference operating point, the two dominant parameters—read noise and aperture diameter—are swept across their realistic ranges in Figure 5.10, now resolved by surface material.

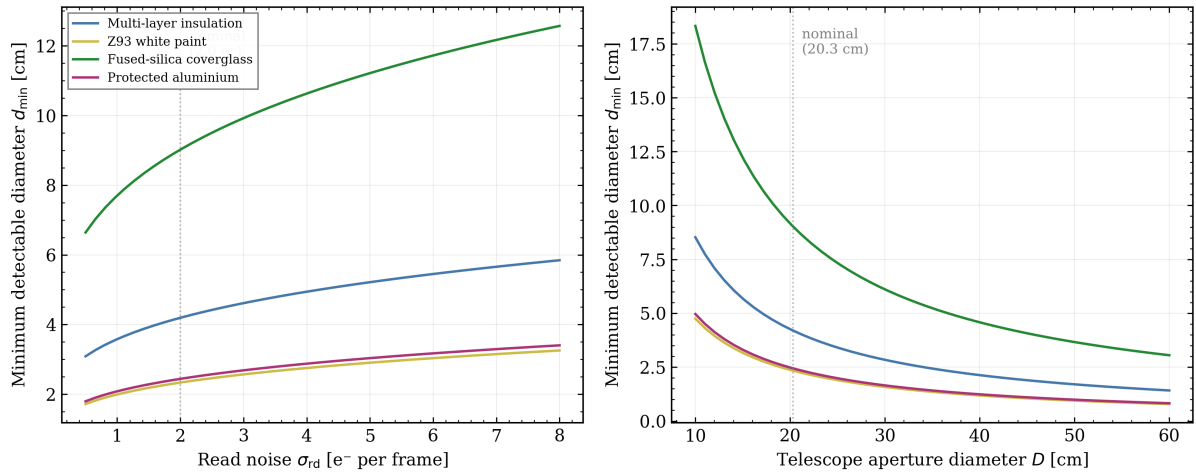


Figure 5.10: Minimum detectable diameter d_{min} for the four candidate surface materials as a function of read noise σ_{rd} (left, at $D = 20.3$ cm and telescope aperture diameter D (right, at $\sigma_{\text{rd}} = 2 e^-$), at the reference geometry ($h = 500$ km, $\epsilon = 30^\circ$, $\alpha = 90^\circ$). All other parameters are held at their reference values; the dotted line marks the nominal operating point.

Discussion

6.1 Principal findings

The framework of Chapter 4 set out to predict photometric detectability for the two target classes from the same photon-counting noise budget, summarized by two figures of merit: the limiting magnitude $m_{\text{lim}}(T_{\text{eff}}, X)$ for astrometric reference stars, and the minimum detectable effective cross-section $A_{\text{eff},\text{min}}$ for LEO targets. Applied to the $8'' f/6.3$ system with the ZWO ASI432MM camera at the Chilean site, both figures of merit resolve into concrete numerical envelopes across spectral type, surface material, and observing geometry. These predictions, anchored to the literature-anchored throughput and the zero point $ZP = 20.34 \pm 0.22$ mag, are the central output of this work and frame the discussion that follows.

For reference stars, the stacked synthetic-tracking limiting magnitude spans roughly $m_{\text{lim}} \approx 14.7$ for hot stars (B5 V) to ≈ 15.8 for cool star (M2 V) with $h = 500$ km, with cooler types reaching fainter by virtue of the negative color correction incurred through the red-extended system bandpass (§5.1.2). The elevation dependence is mild: each spectral class shows a shallow optimum near $\epsilon \approx 30^\circ$ and a peak-to-trough variation of only ~ 0.15 mag across $\epsilon = 20^\circ\text{--}90^\circ$, as the lengthening transit time and the growing atmospheric extinction at low elevation cancel.

For LEO targets, the four representative materials yield limiting magnitudes of $m_{\text{lim}}^{\text{LEO}} \approx 14.2\text{--}14.5$ at the reference geometry, translating into minimum detectable diameters that range from ≈ 0.73 cm in the best-case geometry (Z93 white paint at $h = 400$ km, $\epsilon = 90^\circ$, $\alpha = 30^\circ$) to ≈ 8.10 cm in the worst case for the same material,

white paint, ($h = 1200$ km, $\epsilon = 20^\circ$, $\alpha = 120^\circ$), a factor of ~ 11 across the surveyed envelope. The sub-centimeter best case is the headline result, though it applies only to a favorable corner of the parameter space and is a prediction rather than a measured limit; the more representative twilight geometry ($\alpha \approx 90^\circ$, mid-elevation) yields detectable diameters of a few centimeters.

The remainder of this chapter interprets these numbers against the space-situational-awareness detection gap motivated in Chapter 2 (§6.2), draws out the operational consequences of the contrasting reference-star and LEO observing optima (§6.3), translates the parameter-sensitivity into concrete upgrade guidance (§6.4), and examines the validity of the modeling assumptions and the systemic uncertainties that bound the predictions (§6.5–§6.6).

6.2 Optical fast photometry and the SSA detection gap

Chapter 2 identified the centimeter-scale fragment population as the central blind spot of operational space situational awareness: objects large enough to inflict catastrophic damage at orbital velocities, yet below the ~ 10 cm floor at which radar networks reliably track in LEO, with statistical models placing more than 10^6 such fragments larger than 1 cm in orbit [ESA Space Debris Office, 2025, Haridim et al., 2026]. The detectability predictions of Chapter 5 directly address this gap. Based on these results, a small ground-based optical system operating in the synthetic-tracking regime is, in principle, sensitive to the upper portion of the cm-scale population that radar misses.

This claim must be qualified on three counts before it can be taken as operationally meaningful. First, the sub-centimeter figure occupies only a narrow corner of the parameter space and degrades by a factor of ~ 11 across the surveyed geometry; the honest single number is the few-centimeter limit at typical twilight conditions, not the best case. Second, the predictions are exactly that: they derive from the first-principles throughput, atmosphere, and reflectance models of Chapter 4 and have not been validated against on-sky satellite photometry, as set out in the scope. Third, and most fundamentally, the optical modality carries constraints that radar does not. Detection requires a sunlit target against an astronomically dark sky, restricting useful observation to twilight windows, and the sensitivity is further gated by seeing, cloud cover,

and the sky-brightness assumption that enters the noise budget. The predicted limits describe what the system can reach when those conditions are met, not a continuous all-weather tracking capability.

A further caveat concerns the target population itself. The four representative surfaces of §4.4.3 — MLI, white paint, coverglass, and protected aluminum — characterize the exposed surfaces of *intact* operational spacecraft, whereas the dangerous cm-scale population is overwhelmingly fragmentation debris. Of the four, protected aluminum is the closest proxy for this population, representing exposed structural elements and the degraded surface of older bodies, and it is among the most detectable by virtue of its broadband-flat, high reflectance. To that extent the predictions are encouraging for the fragment regime. However, real fragments present irregular, tumbling, and partially degraded surfaces whose time-averaged reflectance and phase behavior deviate from the clean Lambertian discs assumed here, so the material-resolved limits should be read as an indicative scale for the fragment population rather than a direct prediction for it. Extending the reflectance library to weathered and composite debris surfaces is a natural refinement.

Properly framed, then, the result is not that optical fast photometry supersedes radar or laser ranging, but that it complements them. All three are subject to the same first-order geographic constraint – a ground station must lie beneath the pass, so any single site samples only a small portion of the LEO population and leaves coverage gaps at unserved latitudes. They differ, however, in how that constraint is relaxed and in what additional limitations each carries. Radar provides all-weather, daylight capable tracking but with biased geographical coverage and at substantial infrastructure cost; SLR provides millimeter ranging to cooperative and a limited set of non-cooperative targets [Pearlman et al., 2019]; optical extends sensitivity to fainter objects from any clear dark site and, uniquely, recovers photometric information about the target. Optical’s coverage limitation is thus less about the cost of replication — a low-cost telescope can be redeployed to fill latitude gaps far more readily than a radar installation — than about meteorological and illumination geometry: it requires a clear, dark sky and a target still in sunlight, which restricts observations to nighttime and, near the solstices, to limited windows at mid-latitudes. The framework developed here stops at the per-stack detection threshold by design, but the same calibrated, time-resolved

measurements that establish detectability are precisely those from which object size, spin state, and surface composition are inferred — the characterizing role that motivates fast photometry in the first place. The contribution of this work is therefore to quantify, from first principles, where a low-cost optical system sits within that complementary picture: which targets it can reach, under what geometry, and how its reach responds to upgrade paths examined in 6.4.

6.3 Operational geometry and reference-star availability

The two target classes impose conflicting demands on observing elevation, and reconciling them is the central operational tension of the system. For LEO targets the minimum detectable cross-section is dominated by the slant-range term $A_{\text{eff,min}} \propto r^2$ (Eq. 4.40), which derives the optimum to zenith: detectability improves monotonically with elevation as the line of sight shortens, and while $\sqrt{N}$ gain available from the longer low-elevation transit fails to compensate for the combined slant-range and extinction penalties. For reference stars, by contrast, the source is at effectively infinite distance, the r^2 term drops out, and only atmospheric extinction and transit time compete; these balance to produce the shallow optimum near $\epsilon \approx 30^\circ$ adopted throughout Chapter 5 as the inter-comparison geometry. The system can observe LEO targets near zenith while retaining essentially full reference-star sensitivity.

The binding constraint on the reference-star side is not depth but availability. The synthetic-tracking science exposure is read out over the narrow $38.9' \times 14.5''$ region of interest, and at high Galactic latitude the expected number of detectable reference stars within that strip falls close to unity (Table 5.3). A plate solution requires at least a handful of well-measured stars, so on the ROI alone the system would be unable to astrometrically register a science frame over much of the accessible sky.

The observing strategy resolves this by separating the astrometric solution from the per-frame registration. The plate solution is established before the science exposure from a full-frame calibration image read out over the entire sensor area, which fixes the field coordinates against the abundant reference population available at full FOV. The relevant yield for the plate solve is therefore the per-FOV column of Table 5.3, which provides a comfortably over-determined solution in every field examined. During

the synthetic-tracking science exposure the requirement on the narrow ROI strip is correspondingly relaxed: rather than supplying a full plate solution, the ROI need only contain a single reference star, which anchors the frame-to-frame registration and prevents the pointing from drifting during the exposure under mount shake, tracking error, or derotator imperfection. The decoupling of the full-FOV plate solve from the single-star ROI anchor is thus not merely a convenience but the feature that makes all-sky operation viable: it shifts the binding requirement from “enough stars in the ROI for a plate solution” to “at least one star in the ROI for registration,” a threshold the per-ROI yields of Table 5.3 meet across the surveyed fields.

The latitude gradient in reference-star density also sets a practical sky-coverage envelope for the system. The implication for observation planning is that LEO passes crossing low-latitude fields are calibrated against reference population, whereas passes over the Galactic poles operate close to the astrometric floor; targeting decisions that have the freedom to do so should favor pass geometries that cross denser fields. This is a soft preference rather than a hard constraint, since the per-FOV margin remains adequate even at high latitude, but it identifies the regime in which reference-star availability, rather than target brightness, becomes the limiting factor on astrometric precision.

6.4 Upgrade paths and design guidance

The single-parameter sensitivities of Table 5.5 translate directly into engineering guidance, because they quantify the candidate hardware upgrades by the detectability gain each delivers per unit of effort. At the reference operating point the ordering is unambiguous: read noise dominates, the collecting area follows, and the throughput and photometric aperture are weaker. The practical implication is that the cheapest leverage on detectability lies in the detector rather than the optics: a lower-read-noise sensor improves the limit far more, per dollar and per unit of integration effort, than recoating mirrors or chasing marginal transmission gains. This is the expected behavior of a read-noise-dominated system, and it confirms that the millisecond-exposure regime that motivates the CMOS detector choice in the first place (§3.1) is also the regime in which detector noise performance is the controlling parameter.

The collecting-area sensitivity, while subdominant to read noise on a per-unit basis, offers the largest absolute gains because aperture can be scaled far beyond the few-electron range available for read noise. The $A_{\text{eff,min}} \propto 1/D^2$ scaling of the read-noise-dominated limit means a 1% increase in aperture diameter reduces the cross-section by 2%, and doubling the aperture from $D = 20.3$ cm to 40.6 cm reduces $A_{\text{eff,min}}$ by 75% — a halving of the minimum detectable diameter. Aperture is therefore the upgrade that most directly extends the system’s reach into the cm-scale population, at the cost of the compactness and low expense that motivate the small-telescope approach. The trade-off frames a clear design choice: the present 8” system is optimized for accessibility and cadence, and a move to a larger aperture would shift it toward depth at the expense of those advantages.

The throughput sensitivity acquires additional structure once the wavelength dependence is resolved. For the broadband-flat materials (aluminum, white paint, coverglass) the sensitivity tracks the solar SED and peaks near 500 nm, so a throughput improvement in the green or red produces comparable gains. For the red-sloped MLI reflectance the sensitivity is suppressed in the blue and enhanced in the near-infrared, so an NIR extended detector delivers a substantially larger gain on MLI targets than a blue-enhanced one would. The same hardware upgrade therefore produces material-dependent improvements in $A_{\text{eff,min}}$, and an upgrade aimed at a particular target population should be matched to that population’s spectral reflectance.

Taken together, the sensitivities define a coherent upgrade strategy. The first-order gain comes from read noise, which is bounded; the largest absolute gain comes from aperture, which is expensive and unbounded but erodes the system’s defining advantages; and the throughput gains are smaller, but uniquely tunable to a target population through the wavelength-resolved sensitivity. All three enter the limiting magnitude through the same noise budget, therefore their effects compound predictably: the framework of §4.4.4 allows any combination to be evaluated along a parameterized upgrade path without re-deriving the detectability calculation so a prospective design can be costed against its predicted reduction in $A_{\text{eff,min}}$ before any hardware is acquired. This predictive capacity — the ability to map a hardware change onto a detectability gain from first principles — is one of the primary motivations for building the framework on physical models rather than empirical fits, as set out in the scope.

6.5 Validity of the modeling assumptions

The detectability predictions rest on a chain of physical models, each introducing simplifications whose direction and magnitude bound the credibility of the result. Four assumptions carry the most weight, and for each it is worth stating not merely that it is approximate but which way it biases the predicted limits.

The reflected-sunlight SED is modeled as a Lambertian, rotation-averaged diffuse reflection (§4.4.2), which excludes specular glints by construction. Real spacecraft surfaces — particularly the flat coverglass of solar panels and polished metallic structures — produce specular flashes that can momentarily brighten a target by several magnitudes as the Sun–target–observer geometry passes through a mirror condition. The time-averaged Lambertian treatment therefore yields a *conservative* limit for these surfaces. The framework consequently under-states the instantaneous detectability of specular targets while remaining a defensible predictor of their phase-averaged behavior.

The target is treated as an unresolved point source whose surface is represented by one of four materials, and its intrinsic brightness is collapsed into a single effective cross-section A_{eff} that conflates projected geometric area with average reflections (§4.4.2). Real objects combine several materials with distinct reflectances and present a projected area that varies with attitude and tumble. The four-material decomposition should therefore be read as spanning the plausible range of surface behaviors rather than as a classification of any individual target. The point-source assumption itself is well justified for the cm-to-decimeter population at the system’s PSF scale; it fails only for large nearby objects such as the ISS, which are explicitly excluded.

On the stellar side, three grid-level choices bound the reference-star analysis. The dwarf–giant distinction is implemented as a binary split at $\log g = 3.5$ rather than as an interpolation in surface gravity; the metallicity is fixed at the solar value; and the ATLAS9 grid floor at $T_{\text{eff}} \approx 3500$ K excludes the coolest M dwarfs, where the model under-predicts molecular opacity and a PHOENIX-class grid would be required. The first two are discussed as systematics in §6.6; the third is a population-completeness limit rather than a bias, removing the reddest dwarfs from the per-star yield but not distorting the corrections computed for the stars that remain.

Finally, the atmosphere is modeled with a standard mid-latitude-summer profile

(§4.1.1), used as a stand-in for the dry, high-altitude Atacama site at 1.525 km. The real site is considerably drier than the mid-latitude-summer profile assumes, with correspondingly weaker water-vapor absorption in the near-infrared bands that the red-extended system bandpass samples. The adopted profile is therefore likely *pessimistic* for the site. A site-specific atmospheric profile is the natural refinement, and one that would shift the predictions in the favorable direction rather than undermine them.

6.6 Systematic uncertainties

The dominant systematic in the entire analysis is the throughput normalization. Rather than deriving the absolute system throughput from on-sky standard-star photometry, it is anchored to published end-to-end measurements for comparable instruments and scaled to a peak of 15%, with a ± 3 percentage point spread reflecting the range of plausible values (§4.1.1). This propagates to a ± 0.22 mag uncertainty on the zero point and on every limiting magnitude derived from it — and, through the exponential dependence of $A_{\text{eff,min}}$ on $m_{\text{lim}}^{\text{LEO}}$ (Eq. 4.40), to a $\sim \pm 22\%$ uncertainty on every predicted cross-section and a $\sim \pm 11\%$ uncertainty on every minimum detectable diameter. This is a deliberate consequence of the scope decision to predict rather than measure: the framework is calibrated against the literature so that upgrade paths can be explored without a calibration campaign, at the cost of an absolute-normalization uncertainty that an on-sky campaign could directly reduce. Determining this normalization against standard stars is therefore the single most valuable next measurement, and it is the one quantity whose improvement would tighten every prediction in this work uniformly.

A direct consistency check on the SED treatment is provided by comparing the two independent routes to the color correction (§5.1.4). The population-weighted mean of the per-star Δm_{color} distribution from the Gaia DR3 sample is -0.191 mag, while the spectral-type-weighted average from the class-representative tabulation is -0.201 mag. This agreement validates the class-representative tabulation as a defensible look-up shortcut and, more importantly, demonstrates that neither the dwarf–giant split nor the fixed-metallicity assumption introduces a population-level bias large enough to invalidate the simpler approach. A discrepancy at this level would have signaled

that the binary classification and/or fixed-metallicity was distorting the population-averaged correction; its absence confirms the two methods are mutually consistent.

The geometric inputs carry their own uncertainties, distinct in character from the system systematics because they reflect knowledge of the target rather than calibration of the instrument. The slant-range and phase-angle sensitivities of §4.4.4 quantify how $A_{\text{eff,min}}$ responds to these quantities: a 10% error in slant range produces a 21% error in cross-section, and near quadrature the phase-angle sensitivity is $\sim 2.7\%$ per degree. In an operational setting these translate uncertainties in the target's orbital state into uncertainties on the inferred size. They also set a floor on how precisely a detection can be converted to a physical cross-section without an accompanying orbit solution. They do not, however, affect the detectability threshold itself, which is fixed by the noise budget and the adopted reference geometry; they govern the interpretation of a detection rather than its feasibility.

Conclusion

This thesis developed a first-principles framework for predicting the photometric detectability of LEO satellites and their astrometric reference stars from a small ground-based optical system, and applied it to a specific instrument: an 8" Schmidt–Cassegrain telescope at $f/6.3$ on an equatorial mount, fitted with an IMX432 CMOS detector, and operated in the millisecond-exposure synthetic-tracking regime from a dark Chilean site. The framework reduces the detection problem for both target classes to a single photon-counting noise budget, specialized to the sidereal stack for reference stars and the satellite-rate stack for LEO targets, and expresses the result through two figures of merit: the limiting magnitude $m_{\text{lim}}(T_{\text{eff}}, X)$ and the minimum detectable effective cross-section $A_{\text{eff},\text{min}}$.

The system was characterized from manufacturer specification, atmospheric radiative-transfer modeling, and literature-anchored normalization rather than from on-sky standard-star photometry. The wavelength-dependent throughput was constructed from the digitized detector quantum efficiency and a StarBright-XLT optical model, extrapolated beyond the manufacturer data range to a bare-aluminum reflectance floor, and scaled to a peak of 15% in line with comparable ground-based instruments. Atmospheric transmission was computed with libRadtran over a grid of elevation angles, and the photometric zero point was evaluated against the CALSPEC Vega spectrum as $ZP = 20.34 \pm 0.22$ mag. The ± 0.22 mag uncertainty, inherited from the throughput normalization, is the dominant systematic in the analysis and propagates to every downstream prediction.

For reference stars, the synthetic-tracking framework yields stacked limiting mag-

nitudes spanning $m_{\text{lim}} \approx 14.7$ for hot stars to ≈ 15.8 for cool stars at $h = 500$ km, with a shallow elevation optimum near $\epsilon \approx 30^\circ$ and cooler types reaching fainter through the negative color correction of the red-extended bandpass. Applied to a Gaia DR3 sample across six representative fields, the per-star detection criterion confirmed that reference-star availability, not depth, is the binding constraint at high Galactic latitude, where the per-frame region of interest comes close to containing fewer than one detectable star. The observing strategy resolves this by deriving the plate solution from a prior full-frame exposure, where reference stars are abundant, and requiring only a single anchor star in the science ROI frame-to-frame registration — a threshold met across all fields examined.

The parameter-sensitivity analysis quantified the available upgrade paths from first principles. Read noise dominates the per-unit sensitivity, with a 1 e^- reduction yielding a 24% reduction in $A_{\text{eff,min}}$; collecting area offers the largest absolute gains, with a doubled aperture reducing $A_{\text{eff,min}}$ by 75% at the cost of the system's defining compactness; and throughput gains are smaller but uniquely tunable to a target population through the material-dependent wavelength-resolved sensitivity, where an NIR-extended detector preferentially improves the red-sloped MLI targets. All three enter the same noise budget, therefore the framework allows any combination to be costed against its predicted detectability gain before hardware is acquired — the main benefit of building on physical models rather than empirical fits.

Several directions would extend this work. The most immediate is on-sky validation: a standard-star photometric campaign would directly reduce the ± 0.22 mag normalization uncertainty that dominates every prediction, and real satellite passes would test the predicted limits against measurement. Beyond calibration, the Lambertian, rotation-averaged reflectance model excludes the specular glints and phase-resolved light curves that carry the richest characterization information, and relaxing it would connect the detection framework developed here to the downstream inference of object size, spin state, and attitude. Extending the reflectance library from intact-spacecraft surfaces to the weathered and composite surfaces of fragmentation debris would sharpen the predictions for the cm-scale population that motivates the work most directly. Finally, a site-specific atmospheric profile in place of the mid-latitude-summer standard would refine the transmission grid, likely in the favorable

direction for the dry Atacama site. Each of these builds on the calibrated, modular framework established here, whose contribution is to quantify — from first principles and across the full parameter space — where a low-cost optical system sits within the complementary landscape of space situational awareness.

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